

DSC 140A

All Tags

All problem types.

Tags in this problem set:

- - Gaussians
 - SVMs
 - bayes classifier
 - bayes error
 - computing loss
 - conditional independence
 - convexity
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 - empirical risk minimization
 - feature maps
 - gradient descent
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 - histogram estimators
 - kernel ridge regression
 - least squares
 - linear and quadratic discriminant analysis
 - linear prediction functions
 - linear regression
 - maximum likelihood
 - maximum likelihood
 - naive bayes
 - nearest neighbor
 - object type
 - perceptrons
 - regularization
 - subgradients

Problem #001

Tags: nearest neighbor

Consider the data set shown below:

x	y
-1	2
-3	3
2	4
4	7
5	7
6	10

What is the predicted value of $\langle y \rangle$ at $\langle x = 3 \rangle$ if the 3-nearest neighbor rule is used?

Solution

6

Problem #002

Tags: linear prediction functions

Suppose a linear prediction rule $\langle H(\vec{x}; \vec{w}) = \text{Aug}(\vec{x}) \cdot \vec{w} \rangle$ is parameterized by the weight vector $\langle \vec{w} = (3, -2, 5, 2)^T \rangle$. Let $\langle \vec{z} = (1, -1, -2)^T \rangle$. What is $\langle H(\vec{z}) \rangle$?

Solution

-8

Problem #003

Tags: linear regression, least squares

Suppose a line of the form $\langle H(x) = w_0 + w_1 x \rangle$ is fit to a data set of points $\langle \{(x_i, y_i)\} \rangle$ in $\langle \mathbb{R}^2 \rangle$ by minimizing the mean squared error.

Let the mean squared error of this predictor with respect to this data set be E_1 .

Next, create a new data set by adding a single new point to the original data set with the property that the new point lies *exactly* on the line $H(x) = w_0 + w_1 x$ that was fit above. Let the mean squared error of H on this new data set be E_2 .

Which of the following is true?

$E_1 < E_2$

$E_1 = E_2$

$E_1 > E_2$

Solution

$E_1 > E_2$

Problem #004

Tags: linear prediction functions

Suppose a linear predictor H_1 is fit on a data set $X = \{(\vec{x}_i, y_i)\}$ of n points by minimizing the mean squared error, where each $\vec{x}_i \in \mathbb{R}^d$.

Let $Z = \{(\vec{z}_i, y_i)\}$ be the data set obtained from the original by standardizing each feature. That is, if a matrix were created with the i th row being $(\vec{z}_i)$, then the mean of each column would be zero, and the variance would be one.

Suppose a linear predictor H_2 is fit on this standardized data by minimizing the mean squared error.

True or False: $H_1(\vec{x}_i) = H_2(\vec{z}_i)$ for each $i = 1, \dots, n$.

Solution

True.

Problem #005

Tags: object type

Part 1)

Let Φ be an $(n \times d)$ design matrix, let λ be a real number, and let I be a $(d \times d)$ identity matrix.

What type of object is $(\Phi^T \Phi + n \lambda I)^{-1}$?

A scalar

A vector in $\mathbb{R}^d$

A vector in $\mathbb{R}^n$

A $d \times d$ matrix

A $n \times n$ matrix

Solution

A $d \times d$ matrix

Part 2)

Let Φ be an $n \times d$ design matrix, and let $\vec{y} \in \mathbb{R}^n$. What type of object is $\Phi^T \vec{y}$?

A scalar

A vector in $\mathbb{R}^d$

A vector in $\mathbb{R}^n$

A $d \times d$ matrix

A $n \times n$ matrix

Solution

A vector in $\mathbb{R}^d$

Part 3)

Let $\vec{w} \in \mathbb{R}^{d+1}$, and for each $i \in \{1, 2, \dots, n\}$ let $\vec{x}_i \in \mathbb{R}^d$ and $y_i \in \mathbb{R}$.

What type of object is:

$$\sum_{i=1}^n \left(\vec{w} \cdot \text{Aug}(\vec{x}_i) - y_i \right)^2?$$

A scalar

A vector in $\mathbb{R}^d$

A vector in $\mathbb{R}^{d+1}$

A vector in $\mathbb{R}^n$

A $d \times d$ matrix

A $n \times n$ matrix

Solution

A scalar

Part 4)

Let $(\vec{w} \in \mathbb{R}^{d+1})$, and for each $(i \in \{1, 2, \dots, n\})$ let $(\vec{x}_i \in \mathbb{R}^d)$ and $(y_i \in \mathbb{R})$. Consider the empirical risk with respect to the square loss of a linear predictor on a data set of (n) points:

$$R(\vec{w}) = \frac{1}{2n} \sum_{i=1}^n (\vec{w} \cdot \vec{x}_i - y_i)^2$$

What type of object is $(\nabla R(\vec{w}))$; that is, the gradient of the risk with respect to the parameter vector $(\vec{w})$?

A scalar

A vector in $(\mathbb{R}^d)$

A vector in $(\mathbb{R}^{d+1})$

A vector in $(\mathbb{R}^n)$

A $(d \times d)$ matrix

A $(n \times n)$ matrix

Solution

A vector in $(\mathbb{R}^{d+1})$

Problem #006

Tags: linear prediction functions

Let $(\vec{x}_1 = (-1, -1)^T)$, $(\vec{x}_2 = (1, 1)^T)$, and $(\vec{x}_3 = (-1, 1)^T)$. Suppose (H) is a linear prediction function, and that $(H(\vec{x}_1) = 2)$ while $(H(\vec{x}_2) = -2)$ and $(H(\vec{x}_3) = 0)$.

Let $(\vec{x}_4 = (1, -1)^T)$. What is $(H(\vec{x}_4))$?

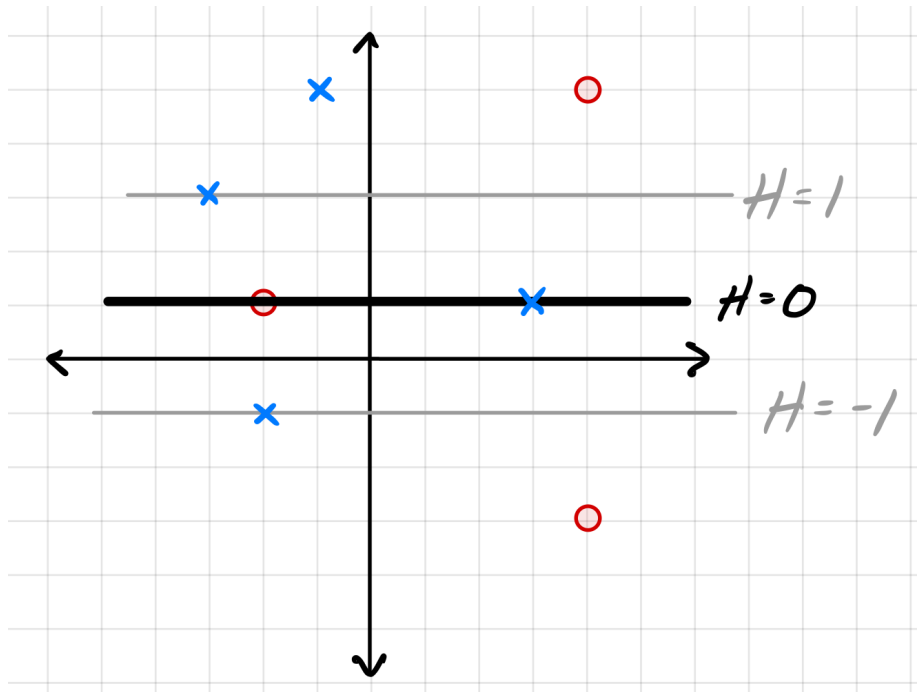
Solution

0

Problem #007

Tags: computing loss

Consider the data set shown below. The “ $(\times)$ ” points have label +1, while the “o” points have label -1. Shown are the places where a linear prediction function (H) is equal to zero, 1, and -1.



For each of the below subproblems, calculate the **total** loss with respect to the given loss function. That is, you should calculate $\sum_{i=1}^n L(\vec{x}_i, y_i, \vec{w})$ using the appropriate loss function in place of L . Note that we have most often calculated the *mean* loss, but here we calculate the *total* so that we encounter fewer fractions.

Part 1)

What is the **total** square loss of (H) on this data set?

Solution

17

Part 2)

What is the **total** perceptron loss of (H) on this data set?

Solution

3

Part 3)

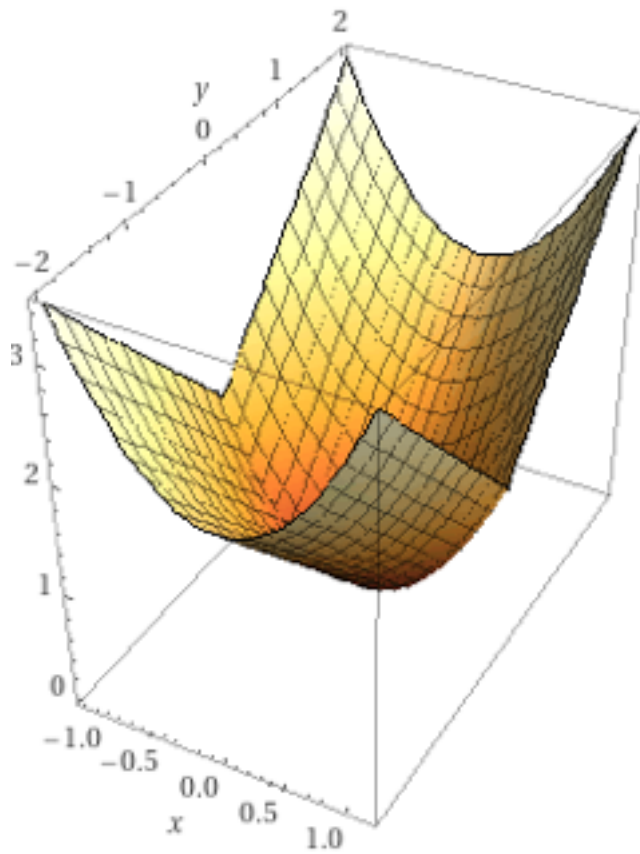
What is the **total** hinge loss of (H) on this data set?

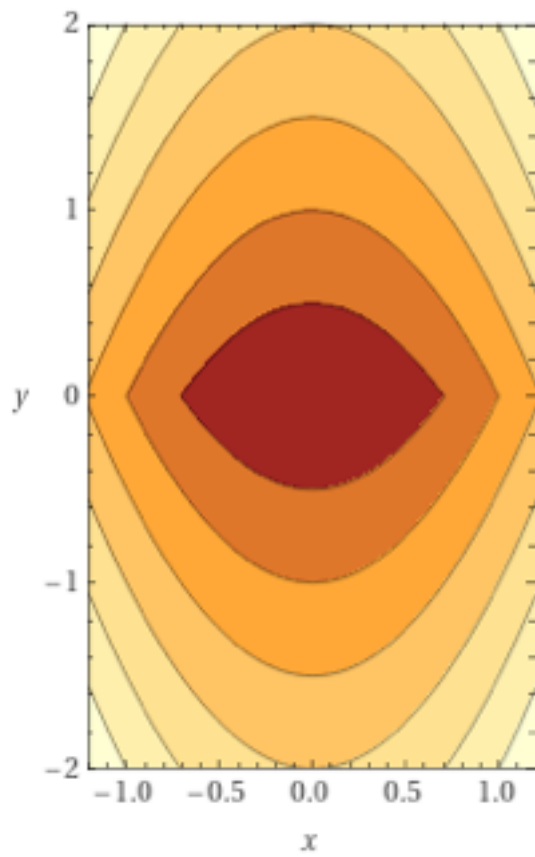
Solution

Problem #008

Tags: subgradients

Consider the function $f(x,y) = x^2 + |y|$. Plots of this function's surface and contours are shown below.





Which of the following are subgradients of f at the point $(0, 0)$? Check all that apply.

$(0, 0)^T$

$(0, 1)^T$

$(0, -1)^T$

$(1, 0)^T$

$(-1, 0)^T$

Solution

$(0, 0)^T$, $(0, 1)^T$, and $(0, -1)^T$ are subgradients of f at $(0, 0)$.

Problem #009

Tags: gradient descent, perceptrons, gradients

Suppose gradient descent is to be used to train a perceptron classifier $(H(\vec{x}; \vec{w}))$ on a data set of (n) points, $(\{\vec{x}_i, y_i\})$. Recall that each iteration of gradient descent takes a step in the opposite direction of the "gradient".

Which gradient is being referred to here?

The gradient of the empirical risk with respect to $(\vec{w})$

The gradient of the empirical risk with respect to $(\{\vec{x}_i\})$

The gradient of the perceptron loss with respect to $(\vec{w})$

The gradient of the perceptron loss with respect to $(\{\vec{x}_i\})$

The gradient of (H) with respect to $(\vec{w})$

The gradient of (H) with respect to $(\vec{x})$

Solution

The gradient of the empirical risk with respect to $(\vec{w})$

Problem #010

Tags: convexity

Let $(\{\vec{x}_i\})$ be a set of (n) vectors in $(\mathbb{R}^d)$. Consider the function $(f(\vec{w}) = \sum_{i=1}^n \vec{w} \cdot \vec{x}_i)$, where $(\vec{w} \in \mathbb{R}^d)$.

True or False: (f) is convex as a function of $(\vec{w})$.

Solution

True.

Problem #011

Tags: SVMs

Let $(X = \{\vec{x}_i, y_i\})$ be a data set of (n) points where each $(\vec{x}_i \in \mathbb{R}^d)$.

Let $(Z = \{\vec{z}_i, y_i\})$ be the data set obtained from the original by standardizing each feature. That is, if a matrix were created with the (i) th row being $(\vec{z}_i)$, then the mean of each column would be zero, and the variance would be one.

Suppose that (X) and (Z) are both linearly-separable. Suppose Hard-SVMs (H_1) and (H_2) are trained on (X) and (Z) , respectively.

True or False: $(H_1(\vec{x}_i) = H_2(\vec{z}_i))$ for each $(i = 1, \dots, n)$.

Solution

False.

This is a tricky problem! For an example demonstrating why this is False, see: <https://youtu.be/LSr55vyvfb4>

Problem #012

Tags: least squares

Suppose a data set $\{(\mathbf{x}_i, y_i)\}$ is linearly-separable.

True or false: a least squares classifier trained on this data set is guaranteed to achieve a training error of zero.

True False

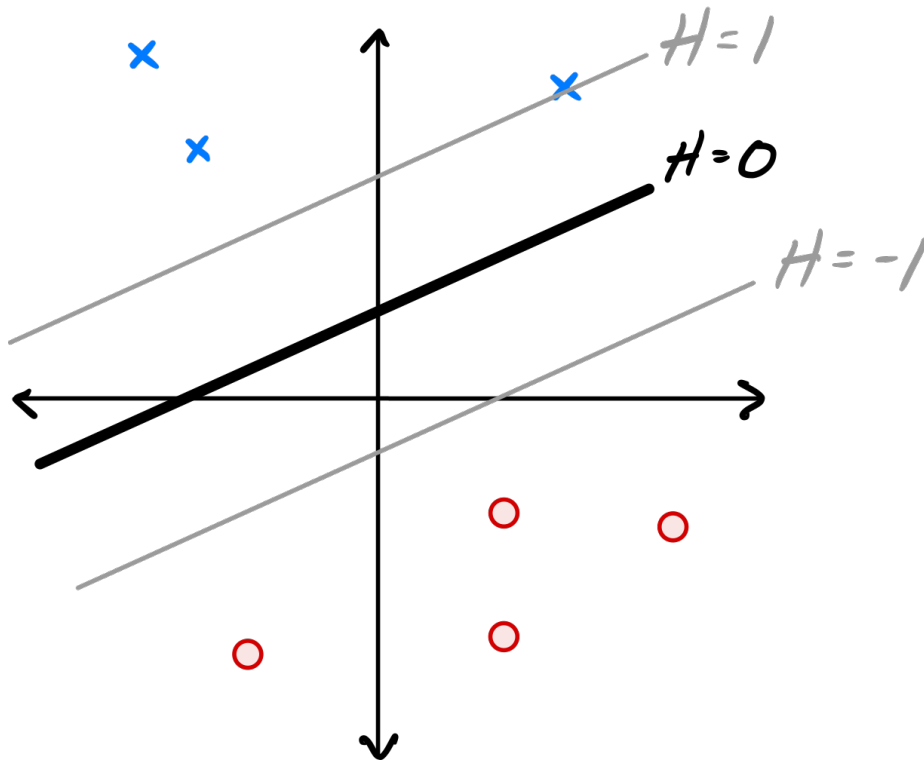
Solution

False

Problem #013

Tags: SVMs

The image below shows a linear prediction function (H) along with a data set; the “ $(\times)$ ” points have label +1 while the “ $(\circ)$ ” points have label -1. Also shown are the places where the output of (H) is 0, 1, and -1.



True or False: $\mathcal{H}$ could have been learned by training a Hard-SVM on this data set.

True False

Solution

False.

The solution to the Hard-SVM problem is a hyperplane that separates the two classes with the maximum margin.

In this solution, the margin is not maximized: There is room for the “exclusion zone” between the two classes to grow, and for the margin to increase.

Problem #014

Tags: kernel ridge regression

Let $\vec{x}_1 = (1, 2, 0)^T$, $\vec{x}_2 = (-1, -1, -1)^T$, $\vec{x}_3 = (2, 2, 0)^T$, $\vec{x}_4 = (0, 2, 0)^T$.

Suppose a prediction function $\mathcal{H}(\vec{x})$ is learned using kernel ridge regression on the above data set using the kernel $\kappa(\vec{x}, \vec{x}') = (1 + \vec{x} \cdot \vec{x}')$.

$\vec{x} \cdot \vec{x}'^2$) and regularization parameter $(\lambda = 3)$. Suppose that $(\vec{\alpha} = (1, 0, -1, 2)^T)$ is the solution of the dual problem.

Let $(\vec{x} = (0, 1, 0)^T)$ be a new point. What is $(H(\vec{x}))$?

Solution

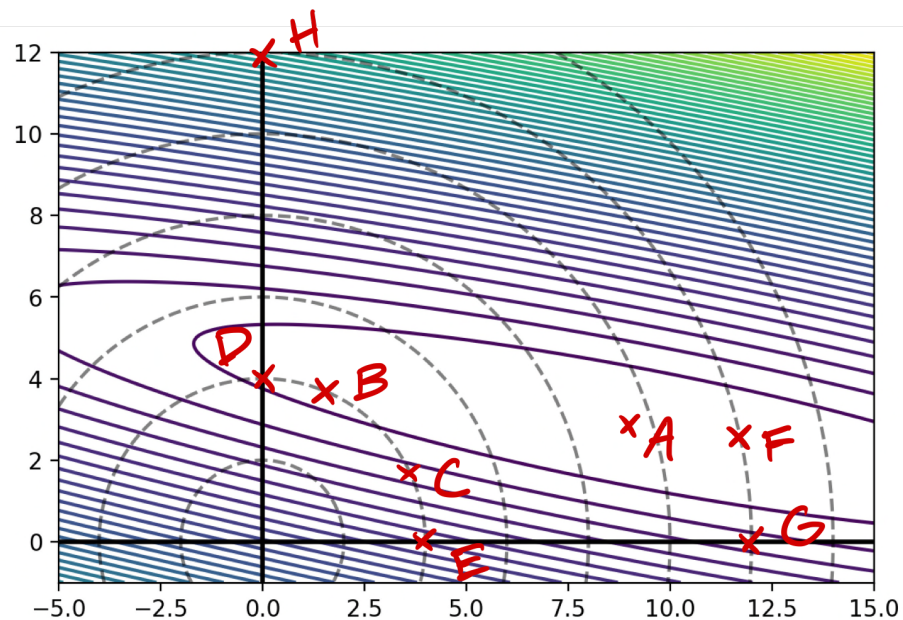
18

Problem #015

Tags: regularization

Let $(R(\vec{w}))$ be the unregularized empirical risk with respect to the square loss (that is, the mean squared error) on a data set.

The image below shows the contours of $(R(\vec{w}))$. The dashed lines show places where $(\|\vec{w}\|_2^2)$ is 2, 4, 6, etc.



Part 1)

Assuming that one of the points below is the minimizer of the *unregularized* risk, $(R(\vec{w}))$, which could it possibly be?

Solution

A

Part 2)

Let the **regularized** risk $\tilde{R}(\vec{w}) = R(\vec{w}) + \lambda \|\vec{w}\|_2^2$, where $\lambda > 0$.

Assuming that one of the points below is the minimizer of the **regularized** risk, $\tilde{R}(\vec{w})$, which could it possibly be?

Solution

B

Problem #016

Tags: kernel ridge regression

Let $\{(\vec{x}_i, y_i)\}$ be a data set of n points, with each $\vec{x}_i \in \mathbb{R}^d$. Recall that the solution to the kernel ridge regression problem is $\vec{\alpha} = (K + n\lambda I)^{-1} \vec{y}$, where (K) is the kernel matrix, (I) is the identity matrix, $\lambda > 0$ is a regularization parameter, and $\vec{y} = (y_1, \dots, y_n)^T$.

Suppose kernel ridge regression is performed with a kernel k that is a kernel for a feature map $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^k$.

What is the size of the kernel matrix, (K) ?

$(d \times d)$

$(k \times k)$

$(n \times n)$

$(n \times d)$

Solution

The kernel matrix is of size $(n \times n)$.

Problem #017

Tags: gradients

Let $f(\vec{w}) = \vec{a} \cdot \vec{w} + \lambda \|\vec{w}\|^2$, where $\vec{w} \in \mathbb{R}^d$, $\vec{a} \in \mathbb{R}^d$, and $\lambda > 0$.

What is the minimizer of f ? State your answer in terms of $\vec{a}$ and λ .

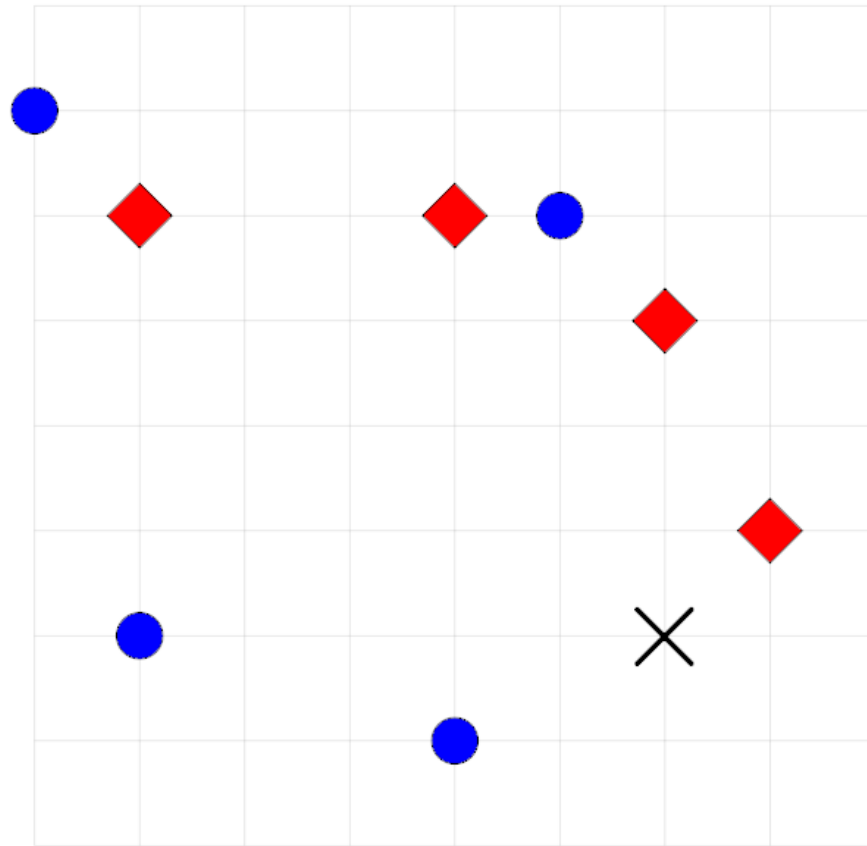
Solution

$$\vec{w}^* = -\frac{\vec{a}}{2\lambda}$$

Problem #018

Tags: nearest neighbor

Consider the data set of diamonds and circles shown below. Suppose a (k) -nn classifier is used to predict the label of the new point marked by $(\times)$, with $(k = 3)$. What will be the prediction? You may assume that the Euclidean distance is used.



○ Class 

○ Class 

Solution

The new point will be classified as a Diamond.

To classify a point with (k) -nn, we examine the (k) (in this case, $(k = 3)$) nearest "neighbors" to whichever point we want to classify. Then, whichever class appears more among those neighbors, we'll predict our new point is part of that class too.

So, what are the three neighbors closest to our new point? Since we assume Euclidean distance (and not some other type of distance, like Manhattan distance), this problem becomes fairly easy to solve from first glance.

Graphically: the three nearest points are the red Diamond to the upper-right of our new point, the blue Circle to the lower-left of our new point, and the red Diamond directly above our new point. The diamond class is more common among the three neighbors, so we predict our new point will also be of the Diamond class.

Mathematically: Euclidean distance is defined by $(D = \sqrt{(\Delta x)^2 + (\Delta y)^2})$, where (Δx) is change in the x-direction and (Δy) is change in the y-direction. We can find the Euclidean distances from each point in our graph to our new point. Whichever three points have the smallest Euclidean distances are the three neighbors we should look at the classes of. After finding all Euclidean distances, we find that the three neighbors we should choose are the same as the ones we chose above, in the visual explanation. The respective Euclidean distances for these three points are $(\sqrt{1^2 + 1^2} = \sqrt{2})$, $(\sqrt{(-1)^2 + (-2)^2} = \sqrt{5})$, and $(\sqrt{0^2 + 3^2} = 3)$, which are shorter than all other distances to the new point in the graph. As an aside, the idea behind (k) -nn is that we assume a property that if two points are similar in location (i.e. they are near each other), then they are likely similar in class. So, (k) -nn is valuable in situations where we have data that is very spatially correlated (ex: predicting price of a new house as "high" or "low" depending on other house prices in the area.)

Solution

To classify a point with (k) -nn, we examine the (k) (in this case, $(k = 3)$) nearest neighbors to whichever point we want to classify. Then, whichever class appears more among those neighbors, we'll predict our new point is part of that class too.

Graphically: The three nearest points are the red Diamond to the upper-right of our new point, the blue Circle to the lower-left of our new point, and the red Diamond directly above our new point. The diamond class is more common among the three neighbors, so we predict our new point will also be of the Diamond class.

Mathematically: Euclidean distance is defined by $(D = \sqrt{(\Delta x)^2 + (\Delta y)^2})$, where (Δx) is change in the x-direction and (Δy) is change in the y-direction. We can find the Euclidean distances from each point in our graph to our new point. Whichever three points have the smallest Euclidean distances to our new point are the three neighbors we should look

at the classes of. After finding all Euclidean distances, we find that the three neighbors we should choose are the same as the ones we chose above, in the visual explanation. The respective Euclidean distances for these three points to our new point are $\sqrt{1^2 + 1^2} = \sqrt{2}$, $\sqrt{(-2)^2 + (-1)^2} = \sqrt{5}$, and $\sqrt{0^2 + 3^2} = 3$, which are shorter than all other distances to the new point in the graph.

Problem #019

Tags: linear prediction functions

Suppose a linear prediction rule $H(\vec{x}; \vec{w}) = \text{Aug}(\vec{x}) \cdot \vec{w}$ is parameterized by the weight vector $\vec{w} = (2, 1, -3, 4)^T$. Let $\vec{z} = (3, -2, 1)^T$. What is $H(\vec{z})$?

Solution

15

Problem #020

Tags: least squares

In the following, let $\mathcal{D}_1$ be a set of points $((x_1, y_1), \dots, (x_n, y_n))$ in $\mathbb{R}^2$. Suppose a straight line $H_1(x) = a_1 x + a_0$ is fit to this data set by minimizing the mean squared error, and let R_1 be the mean squared error of this line.

Now create a second data set, $\mathcal{D}_2$, by doubling the (x) -coordinate of each of the original points, but leaving the (y) -coordinate unchanged. That is, $\mathcal{D}_2$ consists of points $((2x_1, y_1), \dots, (2x_n, y_n))$. Suppose a straight line $H_2(x) = b_1 x + b_0$ is fit to this data set by minimizing the mean squared error, and let R_2 be the mean squared error of this line.

You may assume that all of the (x_i) are unique, as are all of the (y_i) .

Part 1)

Which one of the following is true about R_1 and R_2 ?

$R_1 < R_2$

$R_1 = R_2$

$R_1 > R_2$

Solution

$R_1 = R_2$

Part 2)

Which one of the following is true about a_0 and b_0 (the intercepts)?

$$a_0 < b_0$$

$$a_0 = b_0$$

$$a_0 > b_0$$

Solution

$$a_0 = b_0$$

Part 3)

Which one of the following is true about a_1 and b_1 (the slopes), assuming that the initial slope is positive?

$$a_1 < b_1$$

$$a_1 = b_1$$

$$a_1 > b_1$$

Solution

$$a_1 > b_1$$

Problem #021

Tags: empirical risk minimization

Let D be a data set of points in $\mathbb{R}^d$. Suppose a linear prediction rule $H_1(\vec{x})$ is fit to the data by minimizing the risk with respect to the **square** loss, and suppose another linear prediction rule $H_2(\vec{x})$ is fit to the data by minimizing the risk with respect to the **absolute** loss.

True or False: H_1 and H_2 are guaranteed to be the same; that is, $H_1(\vec{x}) = H_2(\vec{x})$ for all inputs $\vec{x}$.

True False

Solution

False.

Problem #022

Tags: object type

Part 1)

Let $\vec{x} \in \mathbb{R}^d$ and let A be an $(d \times d)$ matrix. What type of object is $\vec{x}^T A \vec{x}$?

A scalar

A vector in $\mathbb{R}^d$

A $(d \times d)$ matrix

Solution

A scalar

Part 2)

Let A be an $(n \times n)$ matrix, and let $\vec{x} \in \mathbb{R}^n$. What type of object is $(A + A^T)^{-1} \vec{x}$?

A scalar

A vector in $\mathbb{R}^n$

An $(n \times n)$ matrix

Solution

A vector in $\mathbb{R}^n$

Part 3)

Suppose we train a support vector machine $H(\vec{x}) = \text{Aug}(\vec{x}) \cdot \vec{w}$ on a data set of (n) points in $\mathbb{R}^d$. What type of object is the resulting parameter vector, $\vec{w}$?

A vector in $\mathbb{R}^d$

A vector in $\mathbb{R}^{d+1}$

A vector in $\mathbb{R}^n$

A vector in $\mathbb{R}^{n+1}$

Solution

A vector in $\mathbb{R}^{d+1}$

Problem #023

Tags: linear prediction functions

Let $\vec{x}_1 = (-1, -1)^T$, $\vec{x}_2 = (1, 1)^T$, and $\vec{x}_3 = (-1, 1)^T$. Suppose H is a linear prediction function, and

that $\langle H(\vec{x}\{1\}) = 2 \rangle$, $\langle H(\vec{x}\{2\}) = -2 \rangle$, and $\langle H(\vec{x}\{3\}) = 2 \rangle$.

Let $\langle \vec{z} = (2, -1)^T \rangle$. What is $\langle H(\vec{z}) \rangle$?

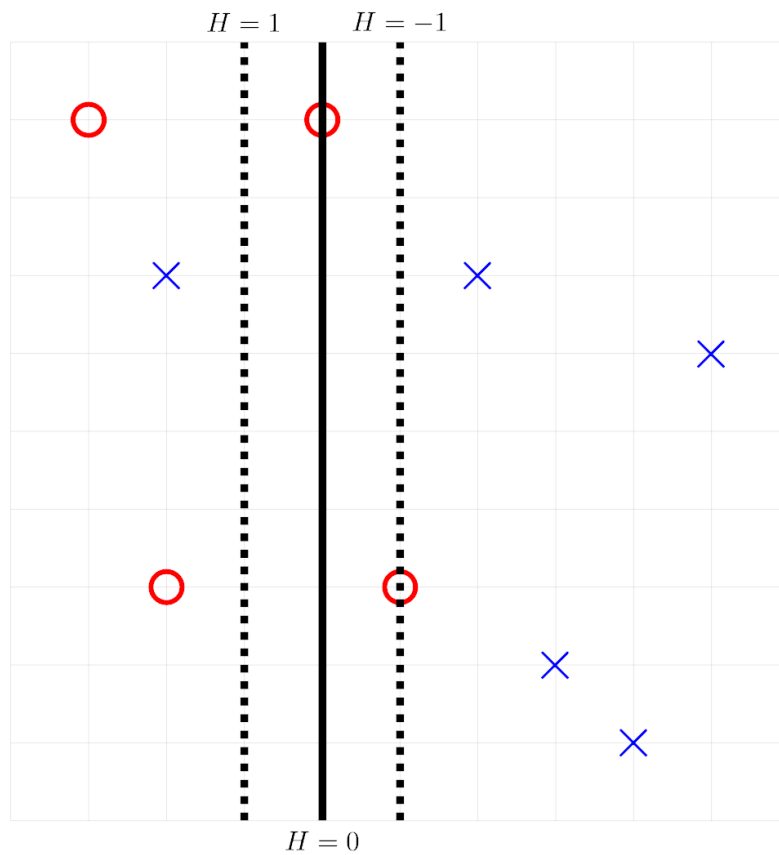
Solution

-4

Problem #024

Tags: computing loss

Consider the data set shown below. The “ $\times$ ” points have label -1, while the “ $\circ$ ” points have label +1. Shown are the places where a linear prediction function $\langle H \rangle$ is equal to zero (the thick solid line), 1, and -1 (the dotted lines). Each cell of the grid is 1 unit by 1 unit.



For each of the below subproblems, calculate the **total** loss with respect to the given loss function. That is, you should calculate $\langle \sum_{i=1}^n$

$L(\vec{x}_i, \vec{w})$ using the appropriate loss function in place of L . Note that we have most often calculated the *mean* loss, but here we calculate the *total* so that we encounter fewer fractions.

Part 1)

What is the **total** square loss of (H) on this data set?

Solution

49

Part 2)

What is the **total** perceptron loss of (H) on this data set?

Solution

3

Part 3)

What is the **total** hinge loss of (H) on this data set?

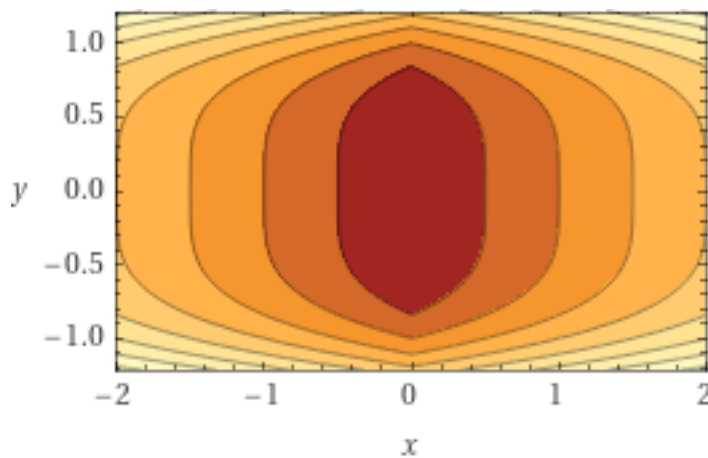
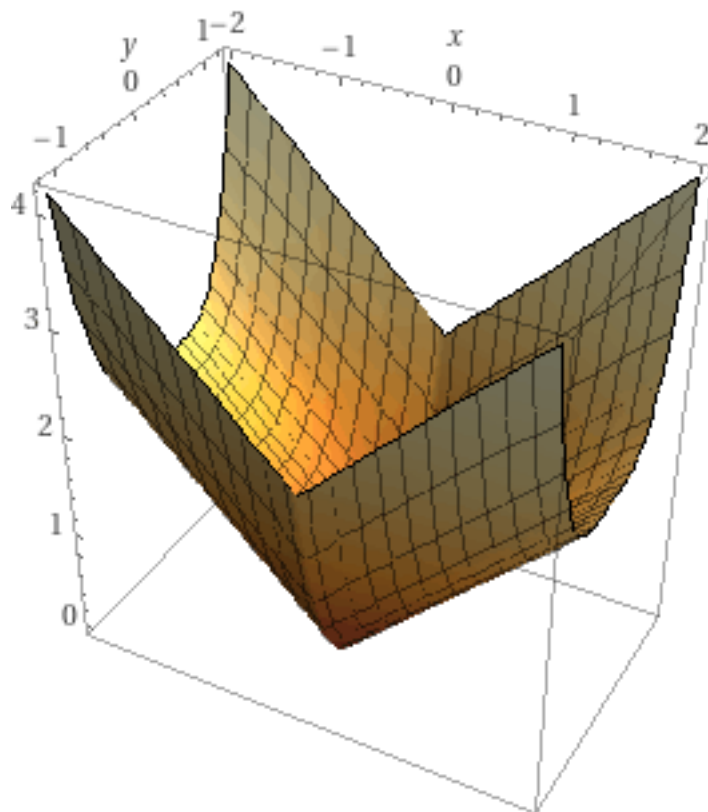
Solution

6

Problem #025

Tags: subgradients

Consider the function $f(x,y) = |x| + y^4$. Plots of this function's surface and contours are shown below.



Write a valid subgradient of f at the point $(0, 1)$.

There are many possibilities, but you need only write one. For the simplicity of grading, please pick a subgradient whose coordinates are both *integers*, and write your answer in the form (a, b) , where a and b are numbers.

Solution

$(x, 4)$, where $x \in [-1, 1]$

Problem #026

Tags: convexity

Consider the function $f(x) = x^4 - x^2$. True or False: f is convex as a function of x .

True False

Solution

False.

Problem #027

Tags: SVMs

Suppose a data set $\mathcal{D}$ is linearly separable. True or False: a Hard-SVM trained on $\mathcal{D}$ is guaranteed to achieve 100% training accuracy.

True False

Solution

True.

Problem #028

Tags: gradient descent

Consider the function $f(x, y) = x^2 + xy + y^2$. Suppose that a single iteration of gradient descent is run on this function with a starting location of $(1, 2)^T$ and a learning rate of $\eta = 1/10$. What will be the x and y coordinates after this iteration?

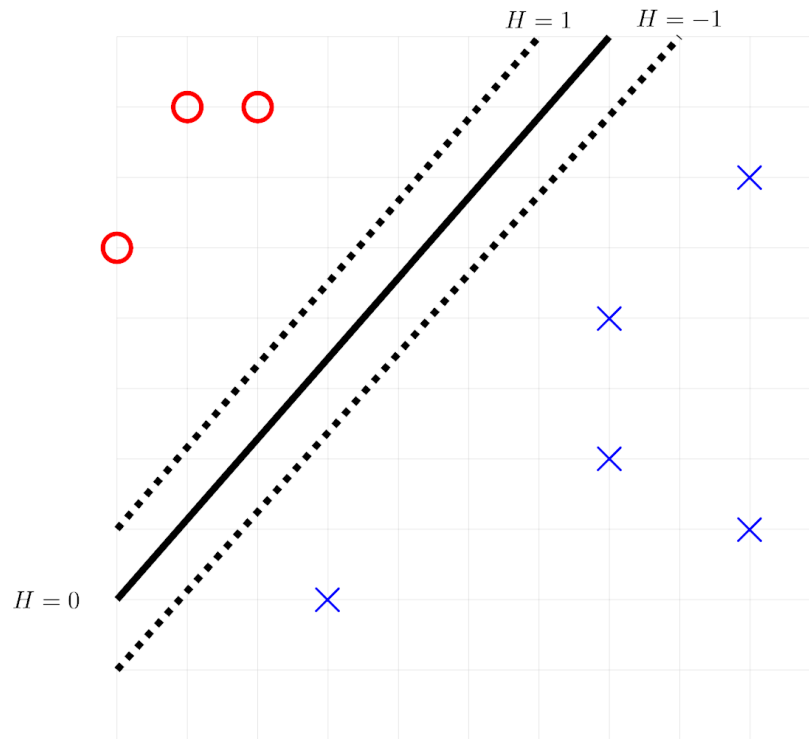
Solution

$(x = 0.6)$, $(y = 1.5)$

Problem #029

Tags: SVMs

The image below shows a linear prediction function H along with a data set; the " $\times$ " points have label +1 while the " $\circ$ " points have label -1. Also shown are the places where the output of H is 0, 1, and -1.



True or False: $\langle H \rangle$ could have been learned by training a Hard-SVM on this data set.

True False

Solution

False.

Problem #030

Tags: kernel ridge regression

Consider the data set: $\langle \vec{x}_1 \rangle = (1, 0, 2)^T$, $\langle \vec{x}_2 \rangle = (-1, 0, -1)^T$, $\langle \vec{x}_3 \rangle = (1, 2, 1)^T$, $\langle \vec{x}_4 \rangle = (1, 1, 0)^T$. Suppose a prediction function $\langle H(\vec{x}) \rangle$ is learned using kernel ridge regression on the above data set using the kernel $\langle \kappa(\vec{x}, \vec{x}') \rangle = (1 + \vec{x} \cdot \vec{x}')^2$ and regularization parameter $\langle \lambda = 3 \rangle$. Suppose that $\langle \vec{\alpha} = (-1, -2, 0, 2)^T \rangle$ is the solution of the dual problem.

Part 1)

What is the (2,3) entry of the kernel matrix?

$\|K_{23}\| = 1$ Show Answer

Part 2)

Let $\vec{x} = (1, 1, 0)^T$ be a new point. What is $H(\vec{x})$?

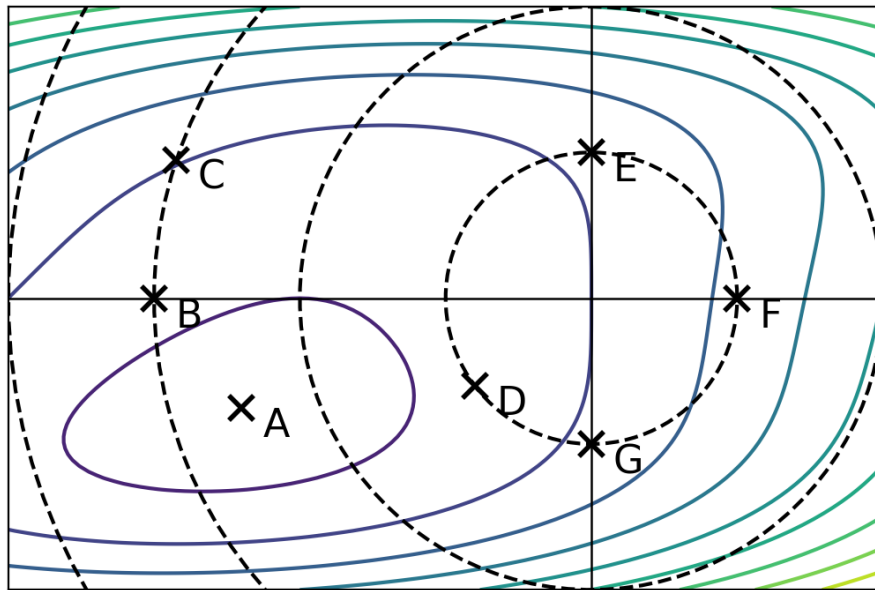
14 Show Answer

Problem #031

Tags: regularization

Let $R(\vec{w})$ be an unregularized empirical risk function with respect to some data set.

The image below shows the contours of $R(\vec{w})$. The dashed lines show places where $\|\vec{w}\|_2$ is 1, 2, 3, etc.



Part 1)

Assuming that one of the points below is the minimizer of the *un*regularized risk, $R(\vec{w})$, which could it possibly be?

- A
- B
- C

D

E

F

G

Solution

A minimizer of the unregularized risk could be point A.

Part 2)

Let the **regularized** risk $\tilde{R}(\vec{w}) = R(\vec{w}) + \lambda \|\vec{w}\|_2^2$, where $(\lambda > 0)$.

Assuming that one of the points below is the minimizer of the **regularized** risk, $\tilde{R}(\vec{w})$, which could it possibly be?

A

B

C

D

E

F

G

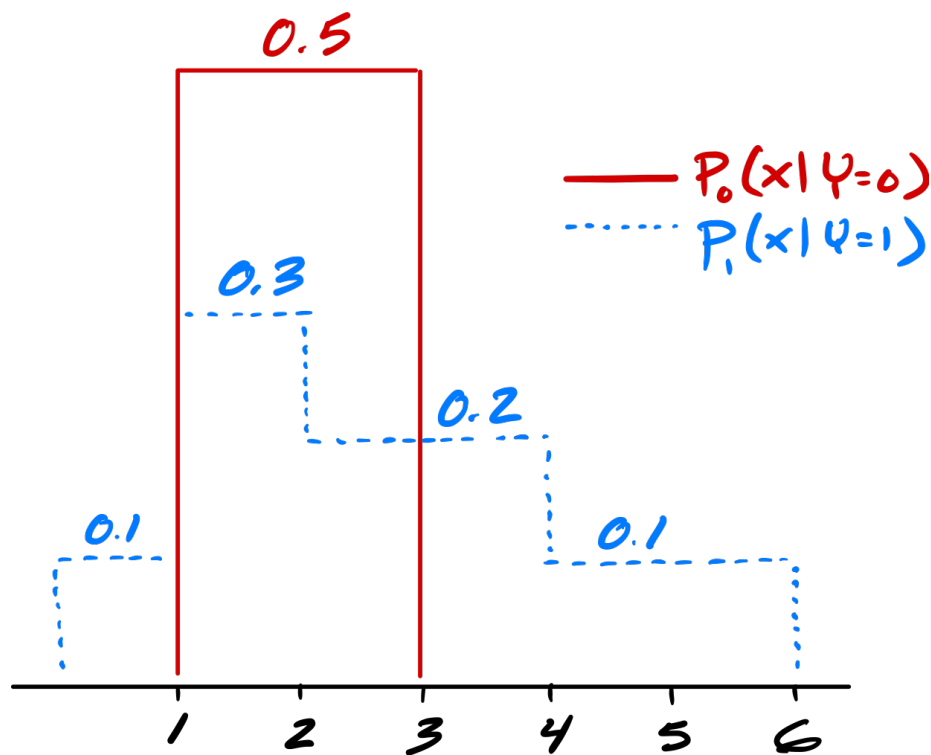
Solution

A minimizer of the regularized risk could be point D.

Problem #032

Tags: bayes error, bayes classifier

Shown below are two conditional densities, $p_1(x|Y=1)$ and $p_0(x|Y=0)$, describing the distribution of a continuous random variable X for two classes: $(Y=0)$ (the solid line) and $(Y=1)$ (the dashed line). You may assume that both densities are piecewise constant.



Part 1)

Suppose $\text{pr}(Y = 1) = 0.5$ and $\text{pr}(Y = 0) = 0.5$. What is the prediction of the Bayes classifier at $x = 1.5$?

Class 0 (the solid class)

Class 1 (the dashed class)

Solution

Class 0

Part 2)

Suppose $\text{pr}(Y = 1) = 0.5$ and $\text{pr}(Y = 0) = 0.5$. What is the Bayes error with respect to this distribution?

0.25 Show Answer

Part 3)

Now suppose $\text{pr}(Y = 1) = 0.7$ and $\text{pr}(Y = 0) = 0.3$. What is the prediction of the Bayes classifier at $x = 1.5$?

Class 0 (the solid class)

Class 1 (the dashed class)

Solution

Class 1

Part 4)

Now suppose $\Pr(Y = 1) = 0.7$ and $\Pr(Y = 0) = 0.3$. What is the Bayes error with respect to this distribution?

0.29 Show Answer

Problem #033

Tags: bayes classifier

Suppose the Bayes classifier achieves an error rate of 15% on a particular data distribution. True or False: It is impossible for any classifier trained on data drawn from this distribution to achieve better than 85% accuracy on a finite test set that is drawn from this distribution.

True False

Solution

False.

Problem #034

Tags: histogram estimators

Consider the data set of ten points shown below:

x
0.3
0.6
1.5
1.9
2.2
2.3
3.5
4.2
6.7
8.1

Suppose this data is used to build a histogram density estimator, $\hat{f}$, with bins: $([0, 2), [2, 6), [6, 10))$. Note that the bins are not evenly sized.

Part 1)

What is $\hat{f}(1.5)$?

0.2 Show Answer

Part 2)

What is $\hat{f}(7)$?

.05 Show Answer

Problem #035

Tags: histogram estimators

Consider this data set of points (x) from two classes $(Y = 1)$ and $(Y = 0)$.

x	y
0.3	0
0.6	0
0.8	1
0.9	0
1.4	0
1.5	1
1.6	1
1.9	1
2.1	0
2.6	0
2.7	1
2.9	0

Suppose a histogram estimator with bins $([0, 1), [1, 2), [2, 3))$ is used to estimate the densities $(p_1(x \text{ given } Y = 1))$ and $(p_0(x \text{ given } Y = 0))$, and these estimates are used in the Bayes classifier to make a prediction.

What will be the predicted class of a new point, $(x = 2.2)$?

Class 0

Class 1

Solution

Class 0.

Problem #036

Tags: density estimation, histogram estimators

Suppose a density estimate $(f: \mathbb{R}^3 \rightarrow \mathbb{R}^1)$ is made using histogram estimators with bins having a length of 2 units, a width of 3 units, and a height of 1 unit.

What is the largest value that $(f(\vec{x}))$ can possibly have?

1/6 Show Answer

Problem #037

Tags: maximum likelihood

Suppose a discrete random variable (X) takes on values of either 0 or 1 and has the distribution:

$$(\text{pr}(X = x) = \theta^x (1 - \theta)^{1-x})$$

where $(\theta \in [0, 1])$ is a parameter.

Given a data set $(x_1, \dots, x_n)$, what is the maximum likelihood estimate for the parameter (θ) ? Show your work.

Solution

The maximum likelihood estimate for (θ) is the mean of all data points: $(\theta = \frac{1}{n} \sum_{i=1}^n x_i)$ We can derive it as follows. First, set up the likelihood function

$$(\text{L}(\theta) = \prod_{i=1}^n \theta^{x_i} (1-\theta)^{1-x_i})$$

Then, to make our lives easier, we can find the log-likelihood before differentiating:

$$(\text{ell}(\theta) = \log \text{L}(\theta) = \sum_{i=1}^n x_i \log \theta + \sum_{i=1}^n (1-x_i) \log(1-\theta))$$

$$(\frac{\partial \text{ell}(\theta)}{\partial \theta} = \frac{\sum_{i=1}^n x_i}{\theta} - \frac{\sum_{i=1}^n (1-x_i)}{1-\theta})$$

$$(\text{Setting } \frac{\partial \text{ell}(\theta)}{\partial \theta} = 0 \rightarrow \frac{\sum_{i=1}^n x_i}{\theta} - \frac{\sum_{i=1}^n (1-x_i)}{1-\theta} = 0)$$

$$(\theta \sum_{i=1}^n (1-x_i) = (1-\theta) \sum_{i=1}^n x_i)$$

$$(\text{and hence } \theta = \frac{1}{n} \sum_{i=1}^n x_i)$$

Problem #038

Tags: covariance

Consider a data set of (n) points in $(\mathbb{R}^d)$, $(\vec{x}_1, \dots, \vec{x}_n)$. Suppose the data are standardized, creating a set of new points $(\vec{z}_1, \dots, \vec{z}_n)$. That is, if the new points are stacked into an $(n \times d)$ matrix, (Z) , the mean and variance of each column of (Z) would be zero and one, respectively.

True or False: the covariance matrix of the standardized data must be the $(d \times d)$ identity matrix; that is, the $(d \times d)$ matrix with ones along the diagonal and zeros off the diagonal.

True False

Solution

False.

Problem #039

Tags: density estimation, maximum likelihood

Suppose data points $(\vec{x}_1, \dots, \vec{x}_n)$ are drawn from an arbitrary, unknown distribution with density (f) .

True or False: it is guaranteed that, given enough data (that is, (n) large enough), a Gaussian fit to the data using the method of maximum likelihood will approximate the true underlying density (f) arbitrarily closely.

True False

Solution

False.

Problem #040

Tags: Gaussians, maximum likelihood

Suppose a Gaussian with a diagonal covariance matrix is fit to 200 points in $(\mathbb{R}^4)$ using the maximum likelihood estimators. How many parameters are estimated? Count each entry of (μ) and the covariance matrix that must be estimated as its own parameter.

8 Show Answer

Problem #041

Tags: covariance

Suppose a data set consists of the following three measurements for each Saturday last year: (X_1) : The day's high temperature (X_2) : The number of people at Pacific Beach on that day (X_3) : The number of people wearing coats on that day

Suppose the covariance between these features is calculated and placed into a (3×3) sample covariance matrix, (C) . Which of the below options most likely shows the sign of each entry of the sample covariance matrix?

$$\begin{pmatrix} + & + & + \\ + & + & + \\ + & + & + \end{pmatrix}$$

$$\begin{pmatrix} + & + & - \\ + & + & - \\ - & - & + \end{pmatrix}$$

$$\begin{pmatrix} + & + & - \\ + & + & - \\ - & - & - \end{pmatrix}$$

$$\begin{pmatrix} - & + & + \\ + & - & + \\ + & + & - \end{pmatrix}$$

$$\begin{pmatrix} - & - & - \\ - & - & - \\ - & - & - \end{pmatrix}$$

Solution

The second option.

Problem #042

Tags: covariance

Suppose we have two data sets, (D_1) and (D_2) , each containing $(n/2)$ points in $(\mathbb{R}^d)$. Let $(\vec{\mu}_1)$ and $(C^{(1)})$ be the mean and sample covariance matrix of (D_1) , and let $(\vec{\mu}_2)$ and $(C^{(2)})$ be the mean and sample covariance matrix of (D_2) .

Suppose the two data sets are combined into a single data set (D) containing (n) points.

Part 1)

True or False: the mean of the combined data (D) is equal to $(\frac{\vec{\mu}_1 + \vec{\mu}_2}{2})$.

True False

Solution

True.

Part 2)

True or False: the sample covariance matrix of the combined data (D) is equal to $(\frac{C^{(1)} + C^{(2)}}{2})$.

True False

Solution

False

Problem #043

Tags: covariance

Suppose a random vector $\vec{X} = (X_1, X_2)$ has a multivariate Gaussian distribution. Suppose it is known that X_1 and X_2 are independent.

Let C be the Gaussian distribution's covariance matrix.

Part 1)

True or False: C must be diagonal.

True False

Solution

True.

Part 2)

True or False: each entry of C must be the same.

True False

Solution

False.

Problem #044

Tags: naive bayes

Consider the below data set which collects information on 10 pets:

Friendly?	Sheds fur?	Species
yes	yes	cat
yes	yes	cat
no	yes	cat
no	yes	cat
no	yes	cat
no	no	cat
yes	no	dog
yes	no	dog
yes	no	dog
yes	yes	dog

Suppose a new pet is friendly and sheds fur. What does a Naïve Bayes classifier predict for its species?

Cat

Dog

Solution

Cat.

Problem #045

Tags: naive bayes

Suppose a data set of 100 points in 10 dimensions is used in a binary classification task; that is, each point is labeled as either a 1 or a 0.

If Gaussian Naive Bayes is trained on this data, how many univariate Gaussians will be fit?

20 Show Answer

Problem #046

Tags: conditional independence

Recall that a deck of 52 cards contains:

Hearts: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

Diamonds: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

Clubs: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

Spades: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

Also recall that Hearts and Diamonds are red, while Clubs and Spades are black.

Part 1)

Suppose a single card is drawn at random.

Let (A) be the event that the card is a heart. Let (B) be the event that the card is a 5.

Are (A) and (B) independent events?

yes, they are independent

no, they are not independent

Solution

Yes, they are independent.

Part 2)

Suppose two cards are drawn at random (without replacing them into the deck).

Let (A) be the event that the second card is a heart. Let (B) be the event that the first card is red.

Are (A) and (B) independent events?

yes, they are independent

no, they are not independent

Solution

No, they are not.

Part 3)

Suppose two cards are drawn at random (without replacing them into the deck).

Let (A) be the event that the second card is a heart. Let (B) be the event that the second card is a diamond. Let (C) be the event that the first card is face card.

Are (A) and (B) conditionally independent given (C) ?

yes, they are conditionally independent

no, they are not conditionally independent

Solution

No, they are not.

Problem #047

Tags: bayes error, bayes classifier

Part 1)

Suppose a particular probability distribution has the property that, whenever data are sampled from the distribution, the sampled data are guaranteed to be linearly separable. True or False: the Bayes error with respect to this distribution is 0%.

True False

Solution

True.

Part 2)

Now consider a different probability distribution. Suppose the Bayes classifier achieves an error rate of 0% on this distribution. True or False: given a finite data set sampled from this distribution, the data must be linearly separable.

True False

Solution

False.

Problem #048

Tags: histogram estimators

Consider this data set of points (x) from two classes $(Y = 1)$ and $(Y = 0)$.

x	y
0.3	0
0.6	0
0.8	1
0.9	0
1.5	1
1.6	1
1.9	1
2.6	0
2.7	1
2.9	0
4.7	1
5.3	1
5.9	0

Suppose a histogram estimator with bins $([0, 2])$, $([2, 4])$, $([4, 6])$ is used to estimate the densities $(p_1(x \text{ given } Y = 1))$ and $(p_0(x \text{ given } Y = 0))$.

What will be the predicted class-conditional density for class 0 at a new point, $(x = 2.2)$? That is, what is the estimated $(p_0(2.2 \text{ given } Y = 0))$?

Solution

1/6.

When estimating the conditional density, we look only at the six points in class zero. Two of these fall into the bin, and the bin width is 2, so the estimated density is:

$$\left[\frac{2}{6} \times 2 = \frac{1}{6}\right]$$

Problem #049

Tags: histogram estimators

Suppose $(\mathcal{D})$ is a data set of 100 points. Suppose a density estimate $(f: \mathbb{R}^3 \rightarrow \mathbb{R}^1)$ is constructed from $(\mathcal{D})$ using histogram estimators with bins having a length of 2 units, a width of 2 units, and a height of 2 units.

The density estimate within a particular bin of the histogram is 0.1. How many data points from $(\mathcal{D})$ fall within that histogram bin?

80 Show Answer

Problem #050

Tags: Gaussians

Suppose data points $(x_1, \dots, x_n)$ are independently drawn from a univariate Gaussian distribution with unknown parameters (μ) and (σ) .

True or False: it is guaranteed that, given enough data (that is, (n) large enough), a univariate Gaussian fit to the data using the method of maximum likelihood will approximate the true underlying density arbitrarily closely.

True False

Solution

True.

Problem #051

Tags: maximum likelihood

Suppose a continuous random variable (X) has the density:

$$[p(x) = \theta e^{-\theta x}]$$

where $(\theta \in (0, \infty))$ is a parameter, and where $(x > 0)$.

Given a data set $(x_1, \dots, x_n)$, what is the maximum likelihood estimate for the parameter (θ) ? Show your work.

Problem #052

Tags: covariance

Let $(\mathcal{D})$ be a set of data points in $(\mathbb{R}^d)$, and let (C) be the sample covariance matrix of $(\mathcal{D})$. Suppose each point in the

data set is shifted in the same direction and by the same amount. That is, suppose there is a vector $\vec{\delta}$ such that if $\vec{x}_i \in \mathcal{D}$, then $\vec{x}_i + \vec{\delta}$ is in the new data set.

True or False: the sample covariance matrix of the new data set is equal to C (the sample covariance matrix of the original data set).

True False

Solution

True.

Problem #053

Tags: Gaussians, maximum likelihood

Suppose a Gaussian with a diagonal covariance matrix is fit to 200 points in $\mathbb{R}^4$ using the maximum likelihood estimators. How many parameters are estimated? Count each entry of $\vec{\mu}$ and the covariance matrix that must be estimated as its own parameter (the off-diagonal elements of the covariance are zero, and shouldn't be included in your count).

8 Show Answer

Problem #054

Tags: Gaussians

Let f_1 be a univariate Gaussian density with parameters μ and σ_1 . And let f_2 be a univariate Gaussian density with parameters μ and $\sigma_2 \neq \sigma_1$. That is, f_2 is centered at the same place as f_1 , but with a different variance.

Consider the density $f(x) = \frac{1}{2}(f_1(x) + f_2(x))$; the factor of $1/2$ is a normalization factor which ensures that f integrates to one.

True or False: f must also be a Gaussian density.

True False

Solution

False. The sum of two Gaussian densities is not necessarily a Gaussian density, even if the two Gaussians have the same mean.

If you try adding two Gaussian densities with different variances, you will get:

$$\left[f(x) = \frac{1}{2} \left(\frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{(x-\mu)^2}{2\sigma_1^2}} + \frac{1}{\sqrt{2\pi}\sigma_2} e^{-\frac{(x-\mu)^2}{2\sigma_2^2}} \right) \right]$$

For this to be a Gaussian, we'd need to be able to write it in the form:

$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$, but this is not possible when $\sigma_1 \neq \sigma_2$.

Problem #055

Tags: covariance

Consider the data set $\mathcal{D}$ shown below.



What will be the sign of the $((1,2))$ entry of the data's sample covariance matrix?

+

-

the entry will be zero

Solution

The sign will be negative.

Problem #056

Tags: linear and quadratic discriminant analysis

Suppose a data set of points in $\mathbb{R}^2$ consists of points from two classes: Class 1 and Class 0. The mean of the points in Class 1 is $((3,0)^T)$, and the mean of points in Class 0 is $((7,0)^T)$. Suppose Linear Discriminant Analysis is performed using the same covariance matrix $(C = \sigma^2 I)$ for both classes, where (σ) is some constant.

Suppose there were 50 points in Class 1 and 100 points in Class 0.

Consider a new point, $((5, 0)^T)$, exactly halfway between the class means. What will LDA predict its label to be?

Class 1

Class 0

There is not enough information to decide

Solution

Class 0.

Problem #057

Tags: naive bayes

Consider the below data set which collects information on the weather on 10 days:

Sunny?	Warm?	Rained?
yes	no	no
yes	yes	no
no	yes	no
no	no	no
yes	no	no
yes	no	no
yes	no	no
no	yes	yes
no	no	yes
no	no	yes

Suppose a new day is not sunny but is warm. What does a Naïve Bayes classifier predict for whether it rained?

yes (it rained)

no (it did not)

Solution

Yes, it rained.

Problem #058

Tags: conditional independence

Suppose that a deck of cards has some cards missing. Namely, both the Ace of Spades and Ace of Clubs are missing, leaving 50 cards remaining.

Hearts: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

Diamonds: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A

Clubs: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K

Spades: 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K

Also recall that Hearts and Diamonds are red, while Clubs and Spades are black.

Part 1)

Suppose a single card is drawn at random.

Let (A) be the event that the card is a heart. Let (B) be the event that the card is an Ace.

Are (A) and (B) independent events?

yes, they are independent

no, they are not independent

Solution

No, they are not.

Part 2)

Suppose a single card is drawn at random.

Let (A) be the event that the card is a red. Let (B) be the event that the card is a heart. Let (C) be the event that the card is an ace.

Are (A) and (B) conditionally independent given (C) ?

yes, they are conditionally independent

no, they are not conditionally independent

Solution

Yes, they are.

Part 3)

Suppose a single card is drawn at random.

Let (A) be the event that the card is a King. Let (B) be the event that the card is red. Let (C) be the event that the card is *not* a numbered card (that is, it is a J, Q, K, or A).

Are (A) and (B) conditionally independent given (C) ?

yes, they are conditionally independent

no, they are not conditionally independent

Solution

No, they are not conditionally independent.

Problem #059

Tags: nearest neighbor

Let $\mathcal{X} = \{(\vec{x}_1, y_1), \dots, (\vec{x}_n, y_n)\}$ be a labeled dataset, where $\vec{x}_i \in \mathbb{R}^d$ is a feature vector and $y_i \in \{-1, 1\}$ is a binary label. Let $\vec{x}$ be a new point that is not in the data set. Suppose a nearest neighbor classifier is used to predict the label of $\vec{x}$, and the resulting prediction is -1 . (You may assume that there is a unique nearest neighbor of $\vec{x}$.)

Now let $\mathcal{Z}$ be a new dataset obtained from $\mathcal{X}$ by scaling each feature by a factor of 2. That is, $\mathcal{Z} = \{(\vec{z}_1, y_1), \dots, (\vec{z}_n, y_n)\}$, where $\vec{z}_i = 2\vec{x}_i$ for each i . Let $\vec{z} = 2\vec{x}$. Suppose a nearest neighbor classifier trained on $\mathcal{Z}$ is used to predict the label of $\vec{z}$.

True or False: the predicted label of $\vec{z}$ must also be -1 .

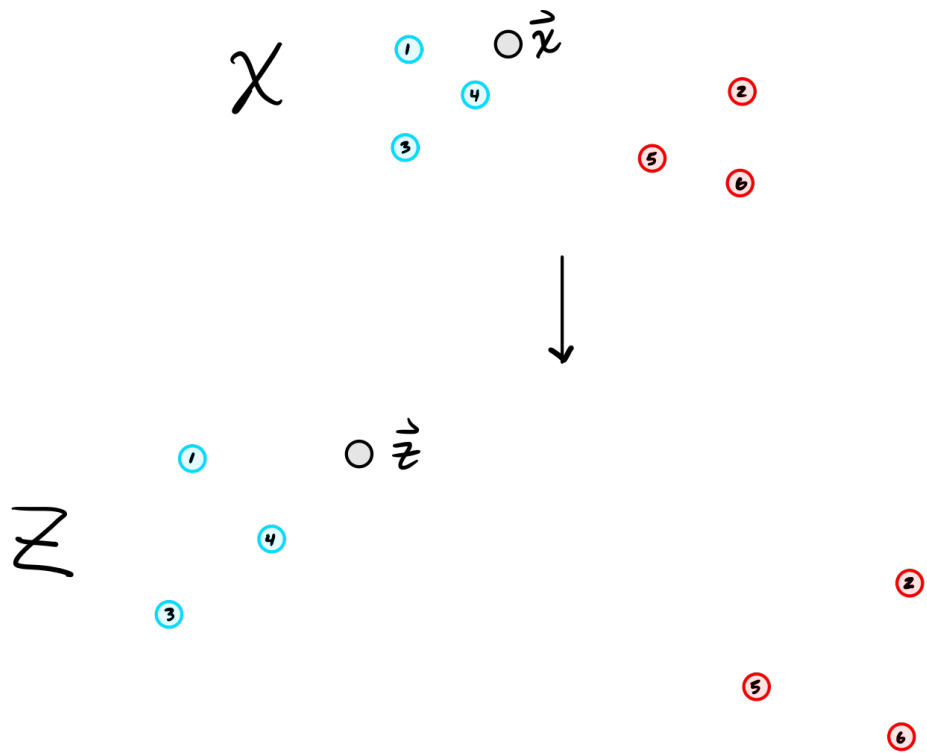
True False

Solution

True.

This question is essentially asking: if we scale a data set by a constant factor, does the nearest neighbor of a point change? The answer is no: therefore, the predicted label cannot change.

When we multiply each feature of a point by the same constant factor, it has the effect of scaling the data set. That is, if $\mathcal{X}$ is the data shown in the top of the image shown below, then $\mathcal{Z}$ is the data shown in the bottom of the image (not drawn to scale):



You can see that the nearest neighbor of $(\vec{x})$ in $(\mathcal{X})$ is point #4, and the nearest neighbor of $(\vec{z})$ in $(\mathcal{Z})$ remains point #4. As such, the predicted label does not change.

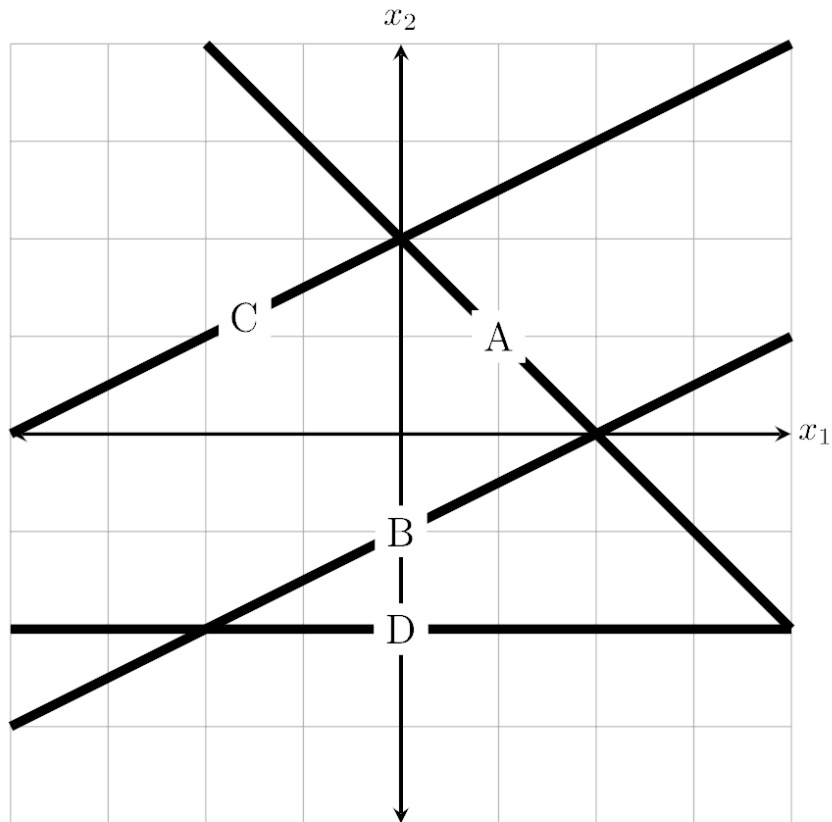
Problem #060

Tags: linear prediction functions

Suppose a linear prediction function $(H(\vec{x}) = w_0 + w_1 x_1 + w_2 x_2)$ is trained to perform classification, and the weight vector is found to be $(\vec{w} = (2, -1, 2)^T)$.

The figure below shows four possible decision boundaries: (A) , (B) , (C) , and (D) . Which of them is the decision boundary of the prediction function (H) ?

You may assume that each grid square is 1 unit on each side, and the axes are drawn to show you where the origin is.



A

B

C

D

Solution

B.

Here are four ways to solve this problem:

Method 1: Guess and check, using the fact that $H(\vec{x})$ must equal zero at every point along the decision boundary. For any of the proposed decision boundaries, we can take a point on the boundary, compute $H(\vec{x})$, and see if it equals 0.

Consider option (A) , and take an arbitrary point on that boundary -- for example, the point $((1, 1)^T)$. Computing $H(\vec{x})$, we get $H(\vec{x})$

$x) = \vec{w} \cdot \text{Aug}(\vec{x}) = (2, -1, 2) \cdot (1, 1, 1) = 3 \neq 0$). Therefore, this cannot be the decision boundary.

Considering option (B), it looks like the point $((0, -1)^T)$ is on the boundary. Computing the value of (H) here, we get $(H(\vec{x}) = \vec{w} \cdot \text{Aug}(\vec{x}) = (2, -1, 2) \cdot (1, -1, 1) = 0)$. That's a good sign, but it's not sufficient to say that this must be the decision boundary (at this point, all we can say is that whatever the decision boundary is, it must go through $((0, -1)^T)$.) To make sure, we pick a second point on the boundary, say, $((2, 0)^T)$. Computing (H) at this point, we get $(H(\vec{x}) = \vec{w} \cdot \text{Aug}(\vec{x}) = (2, -1, 2) \cdot (1, 2, 0) = 0)$. With this, we can conclude that (B) must be the decision boundary.

Method 2: Use the fact that $(\vec{w}_1, \vec{w}_2)$ is orthogonal to the decision boundary. In Homework 02, we learned a useful way of relating the decision boundary to the weight vector $(\vec{w} = (w_0, w_1, w_2)^T)$: it is orthogonal to the vector $(\vec{w}' = (w_1, w_2)^T)$.

In this case, $(\vec{w}' = (-1, 2)^T)$; this is a vector that up and to the left (more up than left). This rules out options (A) and (D), since they can't be orthogonal to that vector. There are several ways to decide between (B) and (C), including the logic from Method 1.

Method 3: Find the equation of the decision boundary in $y = mx + b$ form. Another way to solve this problem is to find the equation of the decision boundary in familiar " $(y = mx + b)$ " form. We can do this by setting $(H(\vec{x}) = 0)$ and solving for (x_2) in terms of (x_1) . We get:

$$\begin{aligned} H(\vec{x}) = 0 &\implies w_0 + w_1 x_1 + w_2 x_2 = 0 \\ &\implies w_2 x_2 = -w_0 - w_1 x_1 \\ &\implies x_2 = -\frac{w_0}{w_2} - \frac{w_1}{w_2} x_1 \end{aligned}$$

Plugging in $(w_0 = 2)$, $(w_1 = -1)$, and $(w_2 = 2)$, we get

$$x_2 = -\frac{2}{2} + \frac{-1}{2} x_1 \implies x_2 = -1 - \frac{1}{2} x_1$$

So the decision boundary is a line with slope $(-\frac{1}{2})$ and (y) -intercept (-1) . This is exactly the equation of the line in option (B).

Method 4: reasoning about the slope in each coordinate. Similar to Method 2, we can rule out options (A) and (D) by reasoning about the slope of the decision boundary in each coordinate.

Imagine this scenario: you're standing on the surface of (H) at a point where $(H = 0)$ (that is, you're on the decision boundary). You will take one step to the right (i.e., one step in the (x_1) direction), which will (in general) cause you to leave the decision boundary and either increase or decrease in height. If you want to get back to the decision boundary, and your next step must be either up or down, which should it be? The answer will tell you whether the decision boundary looks like $(/)$ or like $(\backslash)$.

In this case, taking a step to the right (in the (x_1) direction) will cause you to *decrease* in height. This is because the (w_1) component of $(\vec{w})$ is negative, and so increasing (x_1) will decrease (H) . Now, to get back to the decision boundary, you must *increase* your height to get back to $(H = 0)$. Since the slope in the (x_2) direction is positive (2, to be exact), you must *increase* (x_2) in order to increase (H) . Therefore, the decision boundary is up and to the right: it looks like $(/)$.

Problem #061

Tags: feature maps, linear prediction functions

Suppose a linear prediction function $(H(\vec{x}) = w_1 \phi_1(\vec{x}) + w_2 \phi_2(\vec{x}) + w_3 \phi_3(\vec{x}) + w_4 \phi_4(\vec{x}))$ is fit using the basis functions

$$\phi_1(\vec{x}) = 1, \quad \phi_2(\vec{x}) = x_1^2, \quad \phi_3(\vec{x}) = x_2^2, \quad \phi_4(\vec{x}) = x_1 x_2,$$

where $(\vec{x} = (x_1, x_2)^T)$ is a feature vector in $(\mathbb{R}^2)$. The weight vector $(\vec{w})$ is found to be $(\vec{w} = (1, -2, 3, -4)^T)$.

Let $(\vec{x} = (1, 2)^T)$. What is $(H(\vec{x}))$?

3 Show Answer

Solution

We have:

$$\begin{aligned} H(\vec{x}) &= w_1 \phi_1(\vec{x}) + w_2 \phi_2(\vec{x}) + w_3 \phi_3(\vec{x}) + w_4 \phi_4(\vec{x}) \\ &= 1 \times 1 + (-2) \times x_1^2 + 3 \times x_2^2 + (-4) \times x_1 \times x_2 \\ &= 1 \times 1 + (-2) \times 1^2 + 3 \times 2^2 + (-4) \times 1 \times 2 \\ &= 1 - 2 + 12 - 8 = 3. \end{aligned}$$

Problem #062

Tags: linear regression, least squares

Suppose a linear regression model $(H(\vec{x}) = w_0 + w_1 x_1 + w_2 x_2)$ is trained by minimizing the mean squared error on the data shown below, where (x_1) and (x_2) are the features and (y) is the target:

x_1	x_2	y
2	4	8
3	6	12
4	8	16
5	10	20

True or False: there is a unique weight vector $\vec{w} = (w_0, w_1, w_2)^T$ minimizing the mean squared error for this data set.

True False

Solution

False.

y can be predicted exactly using only x_1 . In particular, $y = 4x_1$. So the weight vector $\vec{w} = (0, 4, 0)^T$ is one that minimizes the mean squared error.

On the other hand, y can also be predicted exactly using only x_2 . In particular, $y = 2x_2$. So the weight vector $\vec{w} = (0, 0, 2)^T$ is another that minimizes the mean squared error.

In fact, there are infinitely many weight vectors that minimize the mean squared error in this case. Geometrically, this is because the data are all along a 2-dimensional line in 3-dimensional space, and by doing regression we are fitting a plane to the data. There are infinitely many planes that pass through a given line, each of them with a different corresponding weight vector, so there are infinitely many weight vectors that minimize the mean squared error.

Now you might be thinking: the mean squared error is convex, so there can be only one minimizer. It is true that the MSE is convex, but that doesn't imply that there is only one minimizer! More specifically, any local minima of a convex function is also a global minimum, but there can be many local minima. Take, for example, the function $f(x, y) = x^2$. The point $(0, 0)$ is a minimizer, but so is $(0, 2)$ and $(0, 100)$, and so on. There are infinitely many minimizers!

Problem #063

Tags: linear prediction functions

Suppose a linear prediction function $\hat{H}(\vec{x}) = w_0 + w_1 x_1 + w_2 x_2$ is fit to data and the following are observed:

At a point, $\vec{x}^{(1)} = (1, 1)^T$, the value of $\hat{H}$ is 4.

At a point, $\vec{x}^{(2)} = (1, 0)^T$, the value of $\hat{H}$ is 0.

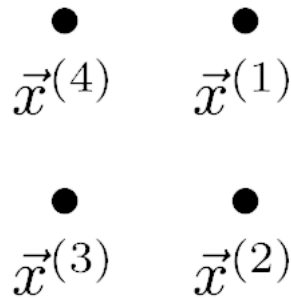
At a point, $\vec{x}^{(3)} = (0, 0)^T$, the value of $\hat{H}$ is -4.

What is the value of $\hat{H}$ at the point $\vec{x}^{(4)} = (0, 1)^T$?

0 Show Answer

Solution

The four points make a square:



Since $\hat{H} = 0$ at $\vec{x}^{(2)}$, the decision boundary must pass through that point. And since $\hat{H}(\vec{x}^{(1)}) = 4 = -\hat{H}(\vec{x}^{(3)})$, the points $\vec{x}^{(1)}$ and $\vec{x}^{(3)}$ are equidistant from the decision boundary. Therefore, the decision boundary must pass exactly halfway between $\vec{x}^{(1)}$ and $\vec{x}^{(3)}$. This makes a line going through $\vec{x}^{(4)}$, meaning that $\vec{x}^{(4)}$ is on the decision boundary, so $\hat{H}(\vec{x}^{(4)}) = 0$.

Problem #064

Tags: object type

Choose the option which best completes the following sentence: In least squares regression, we can fit a linear prediction function $\hat{H}$ by computing the gradient of the _____ with respect to _____ and solving.

square loss, the weights

risk, the weights

square loss, the data

risk, the data

Solution

risk, the weights

Problem #065

Tags: object type

Part 1)

For each $(i = 1, \dots, n)$, let $\vec{x}_i$ be a vector in $\mathbb{R}^d$ and let α_i be a scalar. What type of object is:

$$\sum_{i=1}^n \alpha_i \vec{x}_i?$$

A scalar

A vector in $\mathbb{R}^n$

A vector in $\mathbb{R}^d$

A $(d \times d)$ matrix

An $(n \times n)$ matrix

Solution

A vector in $\mathbb{R}^d$

Part 2)

Let Φ be an $(n \times d)$ matrix, let $\vec{y}$ be a vector in $\mathbb{R}^n$, and let $\vec{\alpha}$ be a vector in $\mathbb{R}^n$. What type of object is:

$$\frac{1}{n} \|\Phi \vec{\alpha} - \vec{y}\|^2 + \vec{\alpha}^T \Phi \Phi^T \vec{\alpha}$$

A scalar

A vector in $\mathbb{R}^n$

A vector in $\mathbb{R}^d$

A $(d \times d)$ matrix

An $(n \times n)$ matrix

Solution

A scalar

Part 3)

Let $\vec{x}$ be a vector in $\mathbb{R}^d$, and let A be a $(d \times d)$ matrix. What type of object is:

$$\frac{\vec{x}^T A \vec{x}}{\vec{x}^T \vec{x}}?$$

A scalar

A vector in $\mathbb{R}^n$

A vector in $\mathbb{R}^d$

A $(d \times d)$ matrix

An $(n \times n)$ matrix

Solution

A scalar

Part 4)

Let A be a $(d \times n)$ matrix. What type of object is $(A A^T)^{-1}$?

A scalar

A vector in $\mathbb{R}^n$

A vector in $\mathbb{R}^d$

A $(d \times d)$ matrix

An $(n \times n)$ matrix

Solution

A $(d \times d)$ matrix

Part 5)

For each $(i = 1, \dots, n)$, let $\vec{x}_i$ be a vector in $\mathbb{R}^d$. What type of object is:

$$\sum_{i=1}^n \vec{x}_i (\vec{x}_i)^T?$$

A scalar

A vector in $\mathbb{R}^n$

A vector in $\mathbb{R}^d$

A $(d \times d)$ matrix

An $(n \times n)$ matrix

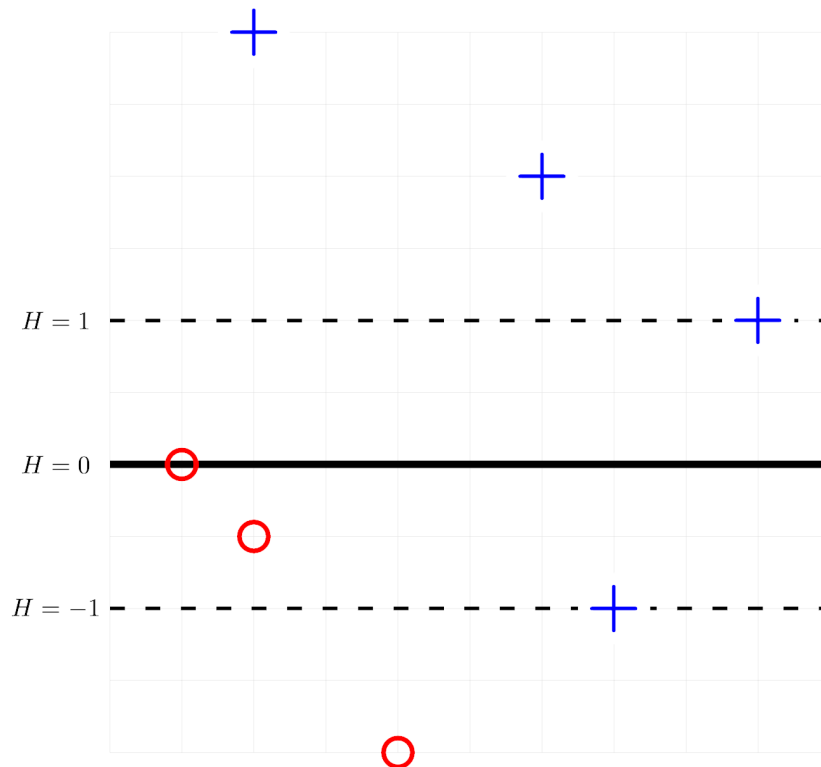
Solution

A $(d \times d)$ matrix

Problem #066

Tags: computing loss

Consider the data set shown below. The 4 points marked with “+” have label +1, while the 3 “○” points have label -1. Shown are the places where a linear prediction function H is equal to zero (the thick solid line), 1, and -1 (the dotted lines). Each cell of the grid is 1 unit by 1 unit. The origin is not specified; it is not necessary to know the absolute coordinates of the data to complete this problem.



For each of the below subproblems, calculate the **total** loss with respect to the given loss function. That is, you should calculate $\sum_{i=1}^n L(\vec{x}_i, y_i, \vec{w})$ using the appropriate loss function in place of L . Note that we have most often calculated the *mean* loss, but here we calculate the *total* so that we encounter fewer fractions.

Part 1)

What is the **total***square* loss of $\backslash(H\backslash)$ on this data set?

11.25 Show Answer

Part 2)

What is the **total***perceptron* loss of $\backslash(H\backslash)$ on this data set?

1 Show Answer

Part 3)

What is the **total***hinge* loss of $\backslash(H\backslash)$ on this data set?

3.5 Show Answer

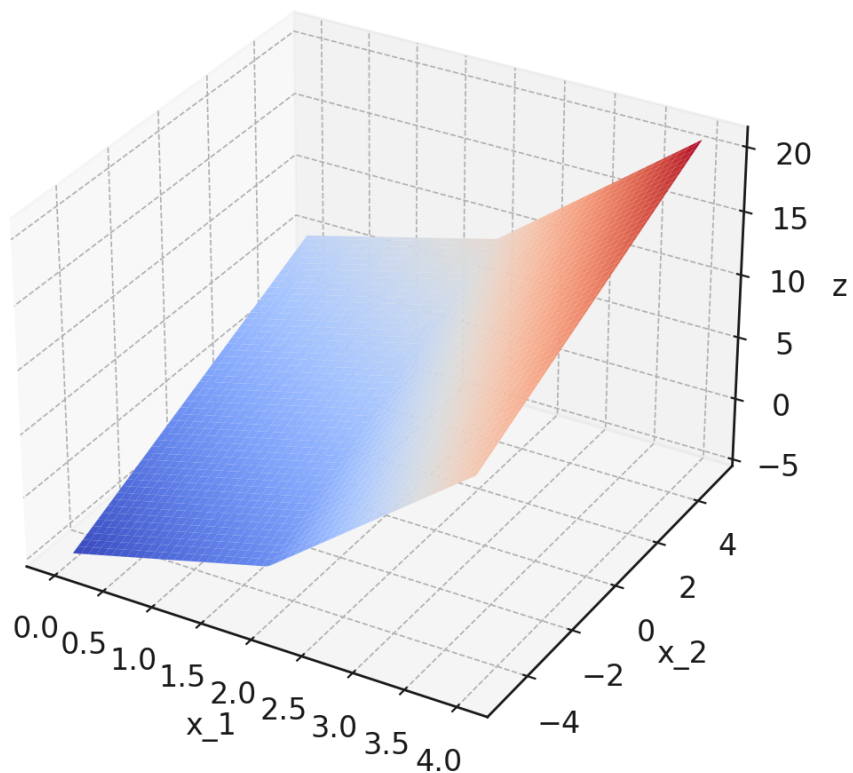
Problem #067

Tags: subgradients

Consider the piecewise function:

$$\backslash[f(x_1, x_2) = \backslashbegin{cases} 2x_1 + x_2 + 8, & \& \backslashtext{if } x_1 < 2, \\ 6x_1 + x_2, & \backslashtext{if } x_1 \geq 2. \end{cases} \backslash]$$

For convenience, a plot of this function is shown below:



Note that the plot isn't intended to be precise enough to read off exact values, but it might help give you a sense of the function's behavior. In principle, you can do this question using only the piecewise definition of f given above.

Part 1)

What is the gradient of this function at the point $((0, 0))$?

$((2, 1))^T$ Show Answer

Part 2)

Which of the below are valid *subgradients* of the function at the point $((2, 0))$? Select all that apply.

$((3, 1))^T$

$((1, 1))^T$

$((6, 2))^T$

$(0, 0)^T$

$(2, 1)^T$

Solution

$(3, 1)^T$ and $(2, 1)^T$ are valid subgradients of the function at $(2, 0)$.

Problem #068

Tags: gradient descent

Recall that a subgradient of the absolute loss is:

$$\begin{cases} \operatorname{Aug}(\vec{x}), & \text{if } \operatorname{Aug}(\vec{x}) \cdot \vec{w} - y > 0, \\ -\operatorname{Aug}(\vec{x}), & \text{if } \operatorname{Aug}(\vec{x}) \cdot \vec{w} - y < 0, \\ 0, & \text{otherwise.} \end{cases}$$

Suppose you are running subgradient descent to minimize the risk with respect to the absolute loss in order to train a function $H(x) = w_0 + w_1 x$ on the following data set:

x	y
1	3
2	5
3	7

Suppose that the initial weight vector is $(\vec{w} = (0, 0)^T)$ and that the learning rate $(\eta = 1)$. What will be the weight vector after one iteration of subgradient descent?

$(1, 2)^T$ Show Answer

Solution

To perform subgradient descent, we need to compute the subgradient of the risk. The main thing to remember is that the subgradient of the risk is the *average* of the subgradient of the loss on each data point.

$\{\operatorname{Aug}\}$

So to start this problem, calculate the subgradient of the loss for each of the three points. Our formula for the subgradient of the absolute loss tells us to compute $\langle \text{Aug}(\vec{x}) \cdot \vec{w} - y \rangle$ for each point and see if this is positive or negative. If it is positive, the subgradient is $\langle \text{Aug}(\vec{x}) \rangle$; if it is negative, the subgradient is $\langle -\text{Aug}(\vec{x}) \rangle$.

Now, the initial weight vector $\langle \vec{w} \rangle$ was conveniently chosen to be $\langle \vec{0} \rangle$, meaning that $\langle \text{Aug}(\vec{x}) \cdot \vec{w} \rangle = 0$ for all of our data points. Therefore, when we compute $\langle \text{Aug}(\vec{x}) \cdot \vec{w} - y \rangle$, we get $\langle -y \rangle$ for every data point, and so we fall into the second case of the subgradient formula for every data point. This means that the subgradient of the loss at each data point is $\langle -\text{Aug}(\vec{x}) \rangle$. Or, more concretely, the subgradient of the loss at each of the three data points is:

$\langle (-1, -1)^T \rangle, \langle (-1, -2)^T \rangle, \langle (-1, -3)^T \rangle$ This means that the subgradient of the risk is the average of these three:

$$\left\langle \frac{1}{3} \left(\begin{bmatrix} -1 \\ -1 \end{bmatrix} + \begin{bmatrix} -1 \\ -2 \end{bmatrix} + \begin{bmatrix} -1 \\ -3 \end{bmatrix} \right) \right\rangle = \left\langle \begin{bmatrix} -1 \\ -2 \end{bmatrix} \right\rangle$$

The subgradient descent update rule says that $\langle \vec{w}^{(1)} \rangle = \langle \vec{w}^{(0)} \rangle - \eta \langle \vec{g} \rangle$, where $\langle \vec{g} \rangle$ is the subgradient of the risk. The learning rate $\langle \eta \rangle$ was given as 1, so we have $\langle \vec{w}^{(1)} \rangle = \langle \vec{w}^{(0)} \rangle - \langle \vec{g} \rangle = \langle \vec{0} \rangle - \langle \begin{bmatrix} -1 \\ -2 \end{bmatrix} \rangle = \langle \begin{bmatrix} 1 \\ 2 \end{bmatrix} \rangle$.

Problem #069

Tags: convexity

Let $\langle \mathcal{X} = \{(\vec{x}_1, y_1), \dots, (\vec{x}_n, y_n)\} \rangle$ be a data set for regression, with each $\langle \vec{x}_i \in \mathbb{R}^d \rangle$ and each $\langle y_i \in \mathbb{R} \rangle$. Consider the following function:

$$\langle R(\vec{w}) = \sum_{i=1}^n |y_i - \vec{w} \cdot \vec{x}_i| + \|\vec{w}\|^2. \rangle$$

True or False: $\langle R(\vec{w}) \rangle$ is a convex function of $\langle \vec{w} \rangle$.

True False

Solution

True.

Problem #070

Tags: regularization

Recall that in ridge regression, we solve the following optimization problem:

$$\min_{\vec{w}} \sum_{i=1}^n (y_i - \vec{w} \cdot \vec{x}_i)^2 + \lambda \|\vec{w}\|^2$$

where $(\lambda > 0)$ is a hyperparameter controlling the strength of regularization.

Suppose you solve the ridge regression problem with $(\lambda = 2)$, and the resulting solution is the weight vector $(\vec{w}_{\text{old}})$. You then solve the ridge regression problem with $(\lambda = 4)$ and find a weight vector $(\vec{w}_{\text{new}})$.

True or False: each component of the new solution, $(\vec{w}_{\text{new}})$, must be less than or equal to the corresponding component of the old solution, $(\vec{w}_{\text{old}})$.

True False

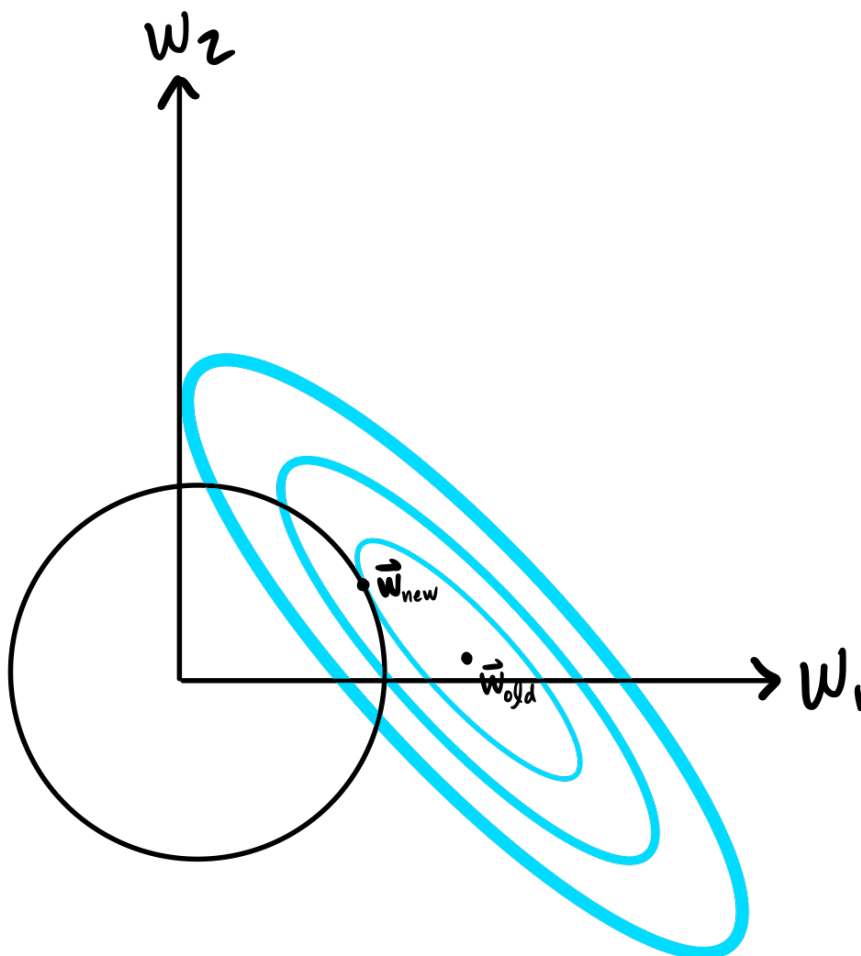
Solution

False.

While it is true that $(\|\vec{w}_{\text{new}}\| \leq \|\vec{w}_{\text{old}}\|)$, this does not imply that each component of $(\vec{w}_{\text{new}})$ is less than or equal to the corresponding component of $(\vec{w}_{\text{old}})$.

The picture to have in mind is that of the contour lines of the mean squared error (which are ovals), along with the circles representing where $(\|\vec{w}\| = c)$ for some constant (c) . The question asked us to consider going from $(\lambda = 2)$ to $(\lambda = 4)$, but to gain an intuition we can think of going from no regularization $(\lambda = 0)$ to some regularization $(\lambda > 0)$; this won't affect the outcome, but will make the story easier to tell.

Consider the situation shown below:



When we had no regularization, the solution was $(\vec{w}_{\text{old}})$, as marked. Suppose we add regularization, and we're told that the regularization is such that when we solve the ridge regression problem, the norm of $(\vec{w}_{\text{new}})$ will be equal to (c) , and that the radius of the circle we've drawn is (c) . Then the solution $(\vec{w}_{\text{new}})$ will be the point marked, since that is the point on the circle that is on the lowest contour.

Notice that the point $(\vec{w}_{\text{new}})$ is closer to the origin, and it's first component is much smaller than the first component of $(\vec{w}_{\text{old}})$. However, the second component of $(\vec{w}_{\text{new}})$ is actually *larger* than the second component of $(\vec{w}_{\text{old}})$.

Problem #071

Tags: SVMs

In lecture, we saw that SVMs minimize the (regularized) risk with respect to the *hinge* loss. Now recall, the 0-1 loss, which was defined in lecture as:

$$L(\vec{x}_i, y_i, \vec{w}) = \begin{cases} 0, & \text{if } y_i = \operatorname{sign}(\vec{w} \cdot \operatorname{Aug}(\vec{x}_i)), \\ 1, & \text{otherwise.} \end{cases}$$

Part 1)

Suppose a **Hard** SVM is trained on linearly-separable data. True or False: the solution is guaranteed to have the smallest 0-1 risk of any possible linear prediction function $(H(\vec{x}) = \vec{w} \cdot \operatorname{Aug}(\vec{x}))$.

True False

Solution

True. If the data is linearly separable, then a Hard-SVM will draw a decision boundary that goes between the two classes, perfectly classifying the training data and achieve a 0-1 risk of zero.

Part 2)

Suppose a **Soft** SVM is trained on linearly-separable data using an unknown slack parameter (C) . True or False: the solution is guaranteed to have the smallest 0-1 risk of any possible linear prediction function $(H(\vec{x}) = \vec{w} \cdot \operatorname{Aug}(\vec{x}))$.

True False

Solution

False.

Part 3)

Suppose a **Soft** SVM is trained on data that is **not** linearly separable using an unknown slack parameter (C) . True or False: the solution is guaranteed to have the smallest 0-1 risk of any possible linear prediction function $(H(\vec{x}) = \vec{w} \cdot \operatorname{Aug}(\vec{x}))$.

True False

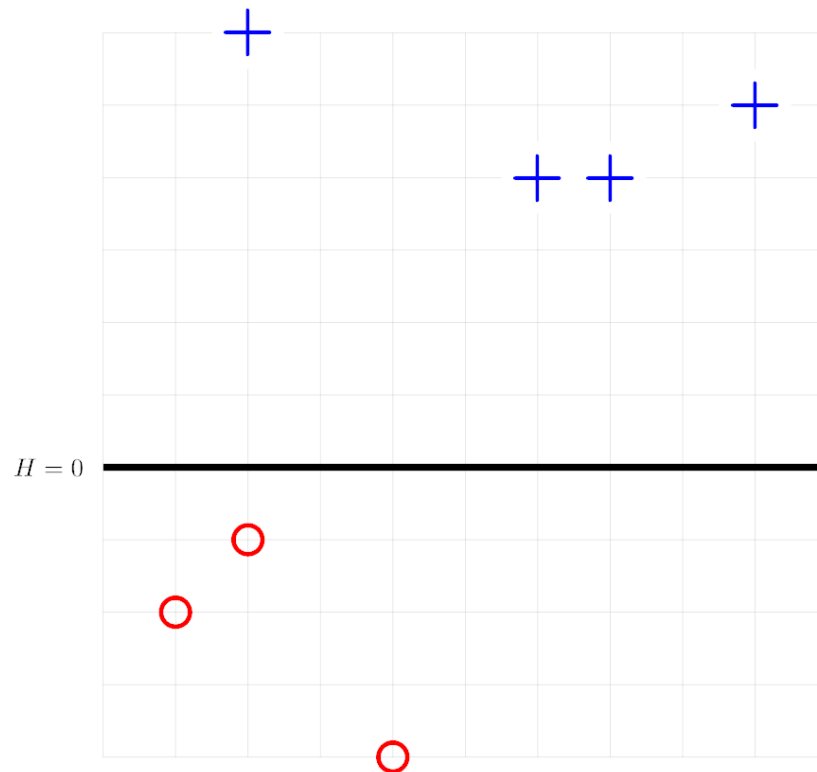
Solution

False.

Problem #072

Tags: perceptrons, gradients

Suppose a perceptron classifier $H(\vec{x}) = \vec{w} \cdot \vec{x}$ is trained on the data shown below. The points marked with “+” have label +1, while the “○” points have label -1. The perceptron's decision boundary is plotted as a thick solid line.



Let $R(\vec{w})$ be the risk with respect to the perceptron loss function, and let $(\vec{w}^*)$ be the weight vector whose corresponding decision boundary is shown above.

True or False: the gradient of $R(\cdot)$ evaluated at $(\vec{w}^*)$ is $(\vec{0})$.

True False

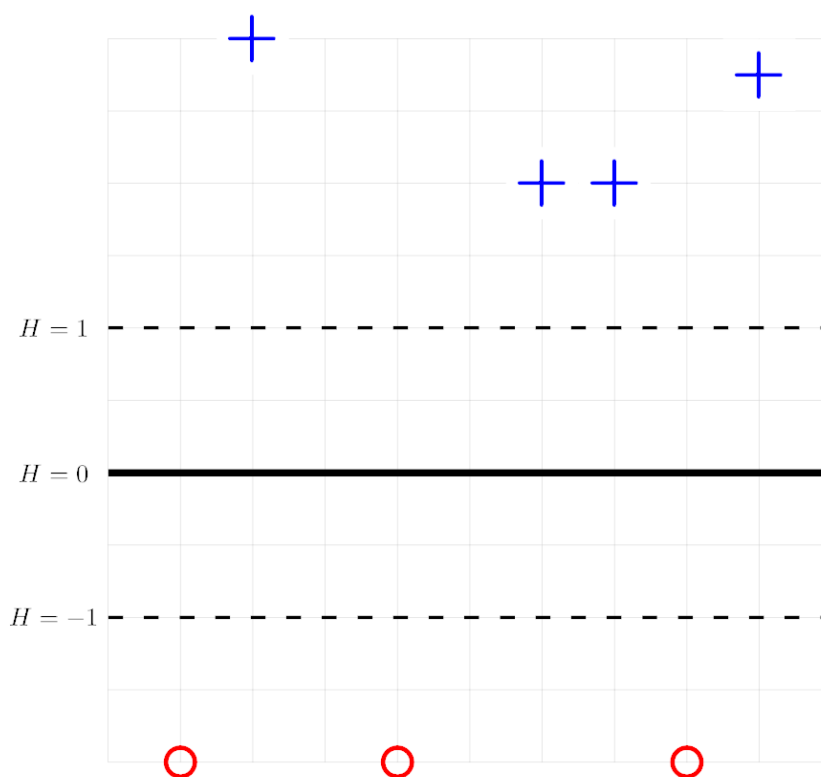
Solution

True.

Problem #073

Tags: SVMs

Consider the data set shown below. The points marked with "+" have label +1, while the "o" points have label -1. Shown are the places where a linear prediction function H is equal to zero (the thick solid line), 1, and -1 (the dotted lines). Each cell of the grid is 1 unit by 1 unit. The origin is not specified (it is not necessary to know the absolute coordinates of any points for this problem).



Suppose that the weight vector $\vec{w}$ of the linear prediction function $(w_0 + w_1 x_1 + w_2 x_2)$ shown above is $((6, 0, \frac{1}{2})^T)$. This weight vector is *not* the solution to the Hard SVM problem.

What weight vector *is* the solution of the Hard SVM problem for this data set?

$((3, 0, \frac{1}{4})^T)$ Show Answer

Solution

Although this decision boundary is in the right place, it can't be the solution to the Hard-SVM problem because its margin isn't maximized. Remember that the surface of (H) is a plane, and in this case the plane is *too steep*; we need to decrease its slope. We do this by scaling the weight vector by a constant factor; in this case, we want to double the margin, so we need to *halve* the slope. Therefore, the right answer is $(\frac{1}{2}(6, 0, \frac{1}{2})^T = (3, 0, \frac{1}{4})^T$.

Problem #074

Tags: gradients

Let $(\vec{x}_1, \dots, \vec{x}_n)$ be vectors in $(\mathbb{R}^d)$ and let $(\vec{w})$ be a vector in $(\mathbb{R}^d)$. Consider the function:

$$f(\vec{w}) = \sum_{i=1}^n \vec{w} \cdot \vec{x}_i + \|\vec{w}\|^2$$

What is $(\frac{d}{d\vec{w}} f(\vec{w}))$?

$(\sum_{i=1}^n \vec{x}_i + 2\vec{w})$ Show Answer

Solution

$$\begin{aligned} \frac{d}{d\vec{w}} \left(\sum_{i=1}^n \vec{w} \cdot \vec{x}_i + \|\vec{w}\|^2 \right) &= \frac{d}{d\vec{w}} \left(\sum_{i=1}^n \vec{w} \cdot \vec{x}_i + \sum_{i=1}^n \vec{w} \cdot \vec{w} \right) \\ &= \sum_{i=1}^n \vec{x}_i + \sum_{i=1}^n \vec{w} \\ &= \sum_{i=1}^n \vec{x}_i + \vec{w} + \vec{w} \\ &= \sum_{i=1}^n \vec{x}_i + 2\vec{w} \end{aligned}$$

Problem #075

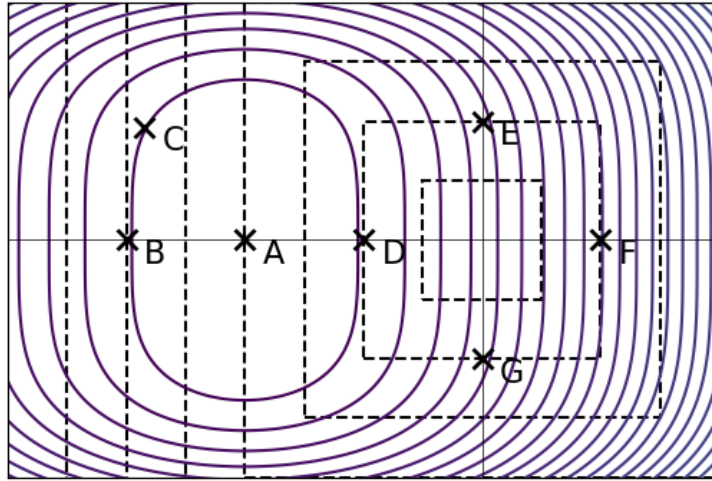
Tags: regularization

The “infinity norm” of a vector $(\vec{w} \in \mathbb{R}^d)$, written $(\|\vec{w}\|_\infty)$, is defined as:

$$\|\vec{w}\|_\infty = \max_{i=1, \dots, d} |w_i|$$

That is, it is the maximum absolute value of any entry of $(\vec{w})$.

Let $(R(\vec{w}))$ be an unregularized risk function on a data set. The solid curves in the plot below are the contours of $(R(\vec{w}))$. The dashed lines show where $(\|\vec{w}\|_\infty)$ is equal to 1, 2, 3, and so on.



Let $\tilde{R}(\vec{w}) = R(\vec{w}) + \lambda \|\vec{w}\|_{\infty}^2$, with $(\lambda > 0)$. The point marked (A) is the minimizer of the unregularized risk. Suppose that it is known that one of the other points is the minimizer of the regularized risk, $\tilde{R}(\vec{w})$, for some unknown $(\lambda > 0)$. Which point is it?

(B)

(C)

(D)

(E)

(F)

(G)

Solution

D.

Problem #076

Tags: object type

Part 1)

Let $f: \mathbb{R}^d \rightarrow \mathbb{R}$ be a function and let $\vec{x}_0$ be a vector in $\mathbb{R}^d$. What type of object is $\frac{d}{d\vec{x}} f(\vec{x}_0)$? In other words, what type of object is the gradient of f evaluated at $\vec{x}_0$?

A scalar

A vector in $\mathbb{R}^n$

A vector in $\mathbb{R}^d$

A $(d \times d)$ matrix

An $(n \times n)$ matrix

Solution

A vector in $\mathbb{R}^d$.

Part 2)

Let Φ be an $(n \times d)$ matrix and let $\vec{\alpha}$ be a vector in $\mathbb{R}^n$. What type of object is:

$$\vec{\alpha}^T \Phi \Phi^T \vec{\alpha}$$

A scalar

A vector in $\mathbb{R}^n$

A vector in $\mathbb{R}^d$

A $(d \times d)$ matrix

An $(n \times n)$ matrix

Solution

A scalar.

Part 3)

For each $(i = 1, \dots, n)$, let $\vec{x}_i$ be a vector in $\mathbb{R}^d$ and y_i be a scalar. Let $\vec{w}$ be a vector in $\mathbb{R}^d$. What type of object is:

$$\frac{1}{n} \sum_{i=1}^n \left(\text{Aug}(\vec{x}_i) \cdot \vec{w} - y_i \right)^2$$

A scalar

A vector in $\mathbb{R}^n$

A vector in $\mathbb{R}^d$

A $(d \times d)$ matrix

An $(n \times n)$ matrix

Solution

A scalar.

Part 4)

Let (X) be an $(n \times d)$ matrix, and assume that $(X^T X)$ is invertible. What type of object is $(X(X^T X)^{-1} X^T)$?

A scalar

A vector in $(\mathbb{R}^n)$

A vector in $(\mathbb{R}^d)$

A $(d \times d)$ matrix

An $(n \times n)$ matrix

Solution

An $(n \times n)$ matrix.

Problem #077

Tags: nearest neighbor

Let $(\mathcal{X} = \{(\vec{x}_1, y_1), \dots, (\vec{x}_n, y_n)\})$ be a labeled dataset, where $(\vec{x}_i \in \mathbb{R}^d)$ is a feature vector and $(y_i \in \{-1, 1\})$ is a binary label. Let $(\vec{x})$ be a new point that is not in the data set. Suppose a nearest neighbor classifier is used to predict the label of $(\vec{x})$, and the resulting prediction is (-1) . (You may assume that there is a unique nearest neighbor of $(\vec{x})$.)

Now let $(\mathcal{Z})$ be a new dataset obtained from $(\mathcal{X})$ by subtracting the same vector $(\vec{\delta})$ from each training point. That is, $(\mathcal{Z} = \{(\vec{z}_1, y_1), \dots, (\vec{z}_n, y_n)\})$, where $(\vec{z}_i = \vec{x}_i - \vec{\delta})$ for each (i) . Let $(\vec{z} = \vec{x} - \vec{\delta})$. Suppose a nearest neighbor classifier trained on $(\mathcal{Z})$ is used to predict the label of $(\vec{z})$.

True or False: the predicted label of $(\vec{z})$ must also be (-1) .

True False

Solution

True.

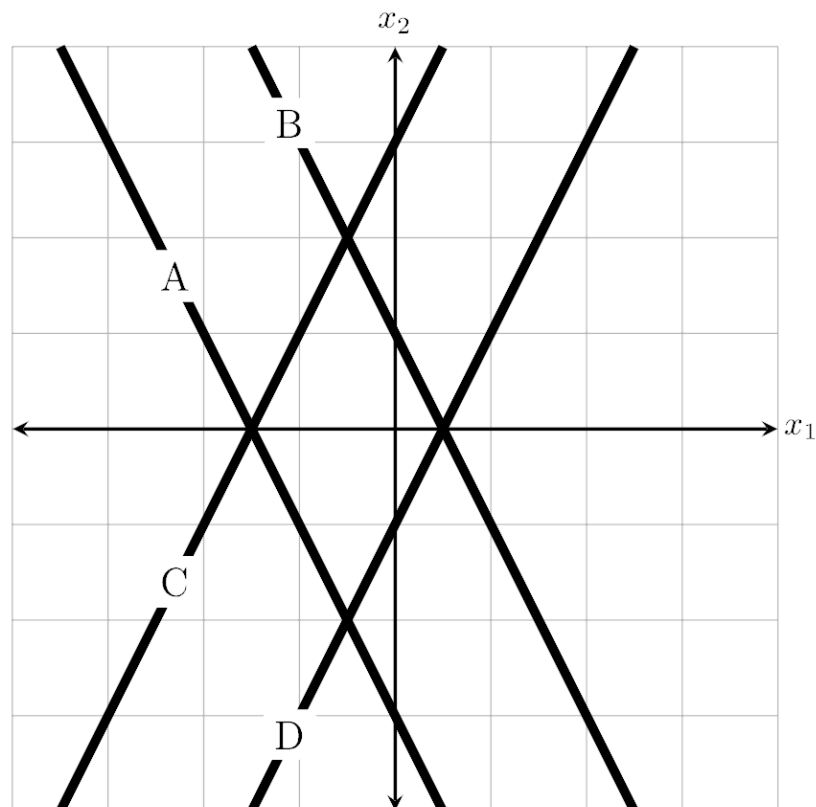
Problem #078

Tags: linear prediction functions

Suppose a linear prediction function $\hat{H}(\vec{x}) = w_0 + w_1 x_1 + w_2 x_2$ is trained to perform classification, and the weight vector is found to be $\vec{w} = (1, -2, 1)^T$.

The figure below shows four possible decision boundaries: (A), (B), (C), and (D). Which of them is the decision boundary of the prediction function $\hat{H}$?

You may assume that each grid square is 1 unit on each side, and the axes are drawn to show you where the origin is.



- A
- B
- C

D

Solution

D.

Problem #079

Tags: feature maps, linear prediction functions

Suppose a linear prediction function $H(\vec{x}) = w_1 \phi_1(\vec{x}) + w_2 \phi_2(\vec{x}) + w_3 \phi_3(\vec{x}) + w_4 \phi_4(\vec{x})$ is fit using the basis functions

$$\phi_1(\vec{x}) = 1, \quad \phi_2(\vec{x}) = x_1 x_2, \quad \phi_3(\vec{x}) = x_2 x_3, \quad \phi_4(\vec{x}) = x_1 x_3,$$

where $\vec{x} = (x_1, x_2, x_3)^T$ is a feature vector in $\mathbb{R}^3$. The weight vector $\vec{w}$ is found to be $\vec{w} = (2, 1, -2, -3)^T$.

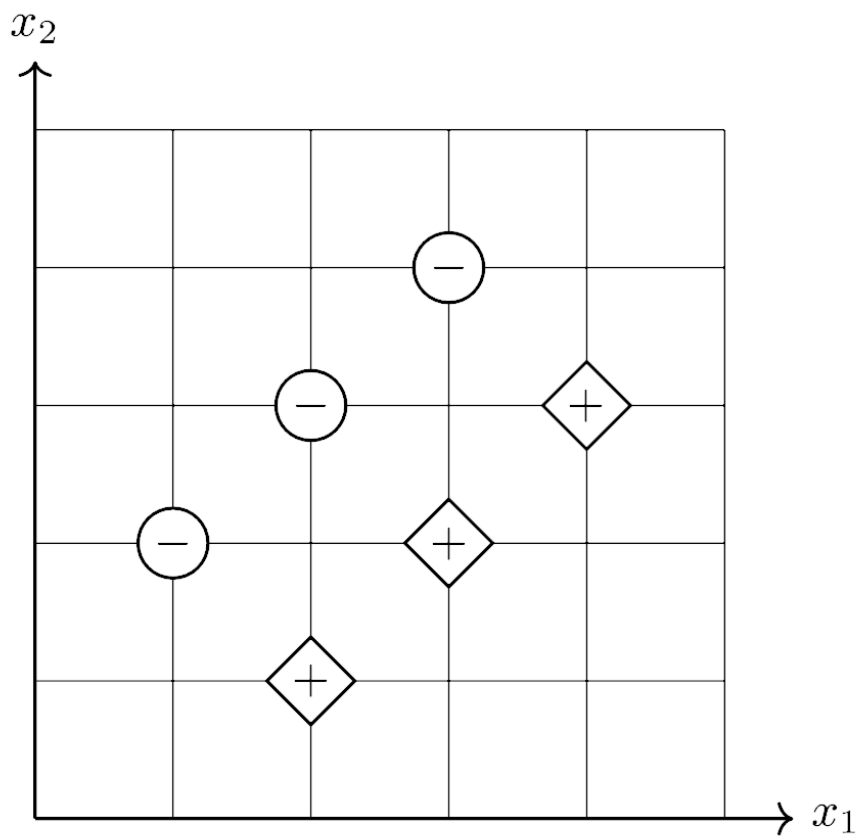
Let $\vec{x} = (1, 2, 1)^T$. What is $H(\vec{x})$?

-3 Show Answer

Problem #080

Tags: least squares, linear prediction functions

Consider the binary classification data set shown below consisting of six points in $\mathbb{R}^2$. Each point has an associated label of either +1 or -1.



What is the mean squared error of a least squares classifier trained on this data set (without regularization)? Your answer should be a number.

0 Show Answer

Problem #081

Tags: convexity

Suppose $f(x_1, x_2)$ is a convex function. Define the new function $g(x) = f(x, 0)$. True or False: $g(x)$ must be convex.

True False

Solution

True.

Problem #082

Tags: convexity

Let $f_1(x)$ be a convex function and let $f_2(x)$ be a non-convex function. Define the new function $f(x) = f_1(x) + f_2(x)$. True or False: $f(x)$ must be non-convex.

True False

Solution

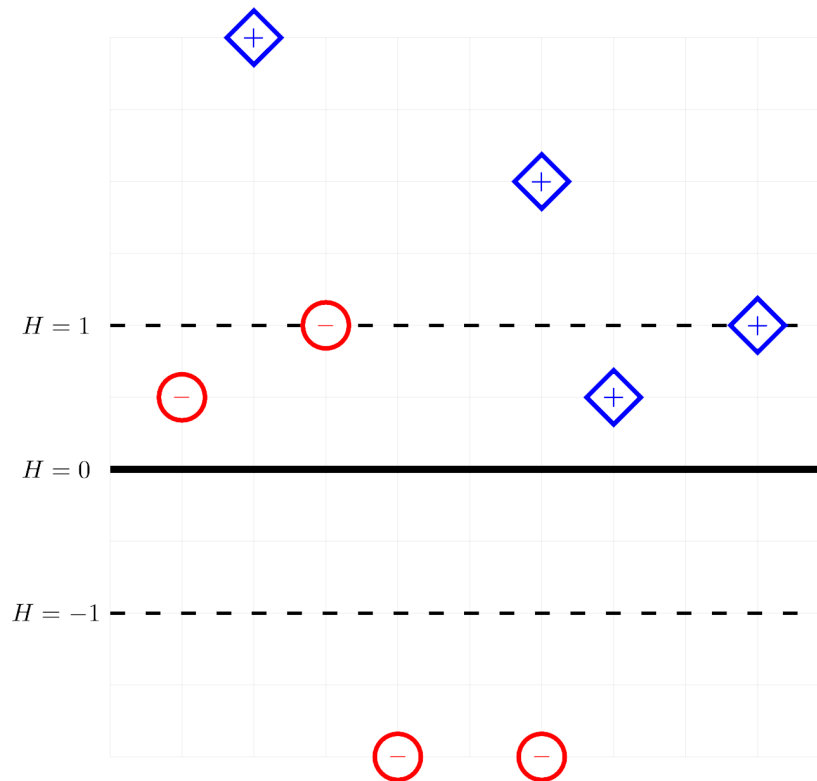
For a simple counterexample, let $f_1(x) = 4x^2$ and $f_2(x) = -x^2$. Then $f(x) = 4x^2 - x^2 = 3x^2$ is convex.

If that example seems too contrived, consider $f_1(x) = x^4 + 3x^2$ and $f_2(x) = x^3$. Then $f(x) = x^4 + x^3 + 3x^2$, and the second derivative test gives us $f''(x) = 12x^2 + 6x + 6$, which is positive for all x .

Problem #083

Tags: computing loss

Consider the data set shown below. The $\diamond$ points marked with “+” have label +1, while the $\circ$ points marked with “-” have label -1. Shown are the places where a linear prediction function H is equal to zero (the thick solid line), 1, and -1 (the dotted lines). Each cell of the grid is 1 unit by 1 unit. The origin is not specified; it is not necessary to know the absolute coordinates of the data to complete this problem.



For each of the below subproblems, calculate the **total** loss with respect to the given loss function. That is, you should calculate $\sum_{i=1}^n L(\vec{x}_i, y_i, \vec{w})$ using the appropriate loss function in place of L . Note that we have most often calculated the *mean* loss, but here we calculate the *total* so that we encounter fewer fractions.

Note: please double-check your calculations! We are not grading your work, so partial credit is difficult to assign on this problem.

Part 1)

What is the **total** *square* loss of (H) on this data set?

13.5 Show Answer

Part 2)

What is the **total** *perceptron* loss of (H) on this data set?

1.5 Show Answer

Part 3)

What is the **totalhinge** loss of (H) on this data set?

4 Show Answer

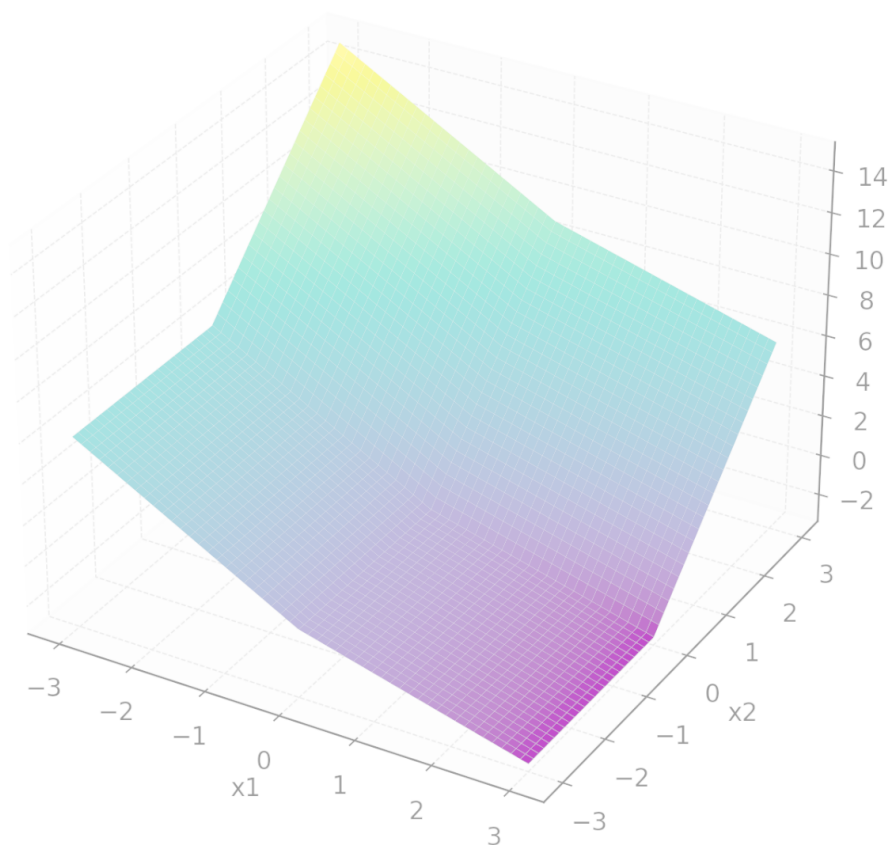
Problem #084

Tags: subgradients, gradients

Consider the following piecewise function that is linear in each quadrant:

```
\[ f(x_1, x_2) = \begin{cases} -2x_1, & \text{if } x_1 < 0 \text{ and } x_2 < 0, \\ -2x_1 + 3x_2, & \text{if } x_1 < 0 \text{ and } x_2 \geq 0, \\ -x_1, & \text{if } x_1 \geq 0 \text{ and } x_2 < 0, \\ -x_1 + 3x_2, & \text{if } x_1 \geq 0 \text{ and } x_2 \geq 0, \end{cases}
```

For convenience, a plot of this function is shown below:



Note that the plot isn't intended to be precise enough to read off exact values,

but it might help give you a sense of the function's behavior. In principle, you can do this question using only the piecewise definition of f given above.

Part 1)

What is the gradient of this function at the point $(1, 1)$?

$(-1, 3)^T$ Show Answer

Part 2)

What is the gradient of this function at the point $(-1, -1)$?

$(-2, 0)^T$ Show Answer

Part 3)

Which of the below are valid *subgradients* of the function at the point $(0, -2)$? Select all that apply.

$(-2, 0)^T$

$(1, 1)^T$

$(-1.5, 0)^T$

$(1, 3)^T$

$(0, 0)^T$

Solution

$(-2, 0)^T$ and $(-1.5, 0)^T$ are valid subgradients.

Part 4)

Which of the below are valid *subgradients* of the function at the point $(0, 0)$? Select all that apply.

$(0, 2)^T$

$(-1, 3)^T$

$(-2, 0)^T$

$(-1, 2)^T$

$(0, 0)^T$

$(0, -1)^T$

Solution

$(-1, 3)^T$, $(-2, 0)^T$, and $(-1, 2)^T$ are valid subgradients.

Problem #085

Tags: gradient descent, subgradients

Recall that a subgradient of the absolute loss is:

$$\begin{cases} \operatorname{Aug}(\vec{x}), & \text{if } \operatorname{Aug}(\vec{x}) \cdot \vec{w} - y > 0, \\ -\operatorname{Aug}(\vec{x}), & \text{if } \operatorname{Aug}(\vec{x}) \cdot \vec{w} - y < 0, \\ \vec{0}, & \text{otherwise.} \end{cases}$$

Suppose you are running subgradient descent to minimize the risk with respect to the absolute loss in order to train a function $H(x) = w_0 + w_1 x$ on the following data set:

x	y
2	2
2	1
5	5

Suppose that the initial weight vector is $(\vec{w} = (0, 0)^T)$ and that the learning rate is $(\eta = 1)$. What will be the weight vector after one iteration of subgradient descent?

$((1, 3)^T)$ Show Answer

Problem #086

Tags: regularization

Recall that in ridge regression, we solve the following optimization problem:

$$\arg\min_{\vec{w}} \sum_{i=1}^n (y_i - \vec{w} \cdot \operatorname{Aug}(\vec{x}_i))^2 + \lambda \|\vec{w}\|^2.$$

where $(\lambda > 0)$ is a hyperparameter controlling the strength of regularization.

Suppose you solve the ridge regression problem with $(\lambda = 2)$, and the resulting solution has a mean squared error of (10) .

Now suppose you increase the regularization strength to $\lambda = 4$ and solve the ridge regression problem again. True or False: it is possible that the mean squared error of the new solution is less than 10 .

By “mean squared error,” we mean $\frac{1}{n} \sum_{i=1}^n (y_i - \vec{w} \cdot \vec{x}_i)^2$

True False

Solution

False.

Problem #087

Tags: SVMs

Let $(\mathcal{X})$ be a binary classification data set $((\vec{x}_1, y_1), \dots, (\vec{x}_n, y_n))$, where each $\vec{x}_i \in \mathbb{R}^d$ and each $y_i \in \{-1, 1\}$. Suppose that the data in $(\mathcal{X})$ is linearly separable.

Let $\vec{w}_{\text{svm}}$ be the weight vector of the hard-margin support vector machine (SVM) trained on $(\mathcal{X})$. Let $\vec{w}_{\text{lsq}}$ be the weight vector of the least squares classifier trained on $(\mathcal{X})$.

True or False: it must be the case that $\vec{w}_{\text{svm}} = \vec{w}_{\text{lsq}}$.

True False

Solution

False.

Problem #088

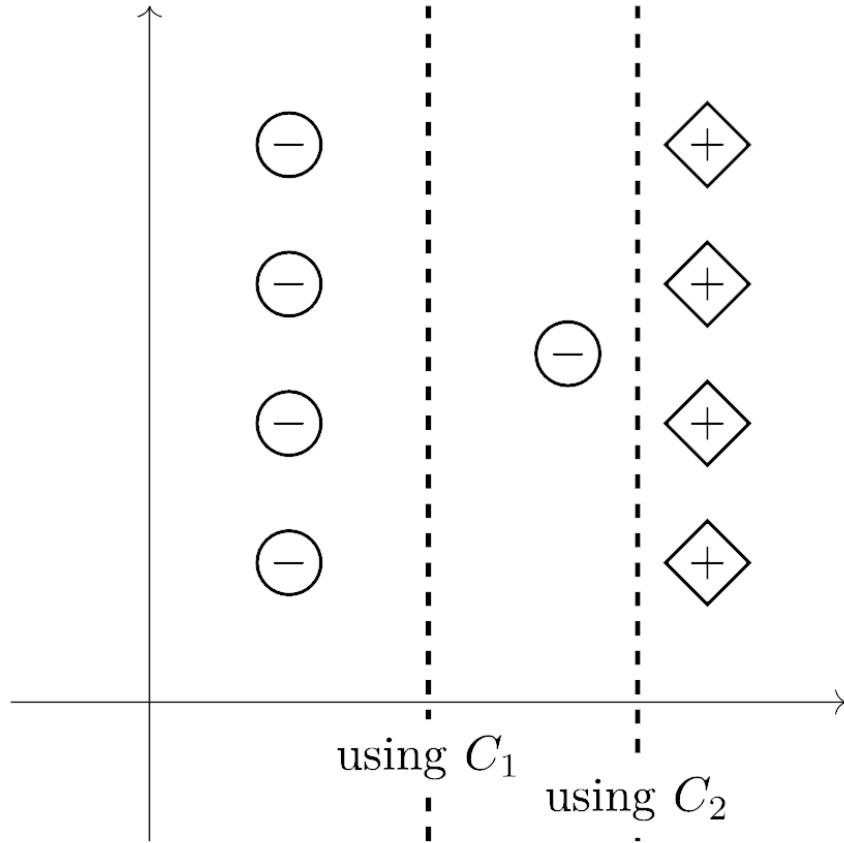
Tags: SVMs

Recall that the Soft-SVM optimization problem is given by

$$\min_{\vec{w} \in \mathbb{R}^{d+1}, \vec{\xi} \in \mathbb{R}^n} \|\vec{w}\|^2 + C \sum_{i=1}^n \xi_i,$$

where $\vec{w}$ is the weight vector, $\vec{\xi}$ is a vector of slack variables, and C is a regularization parameter.

The two pictures below show as dashed lines the decision boundaries that result from training a Soft-SVM on the same data set using two different regularization parameters: C_1 and C_2 .



True or False: $(C_1 < C_2)$.

True False

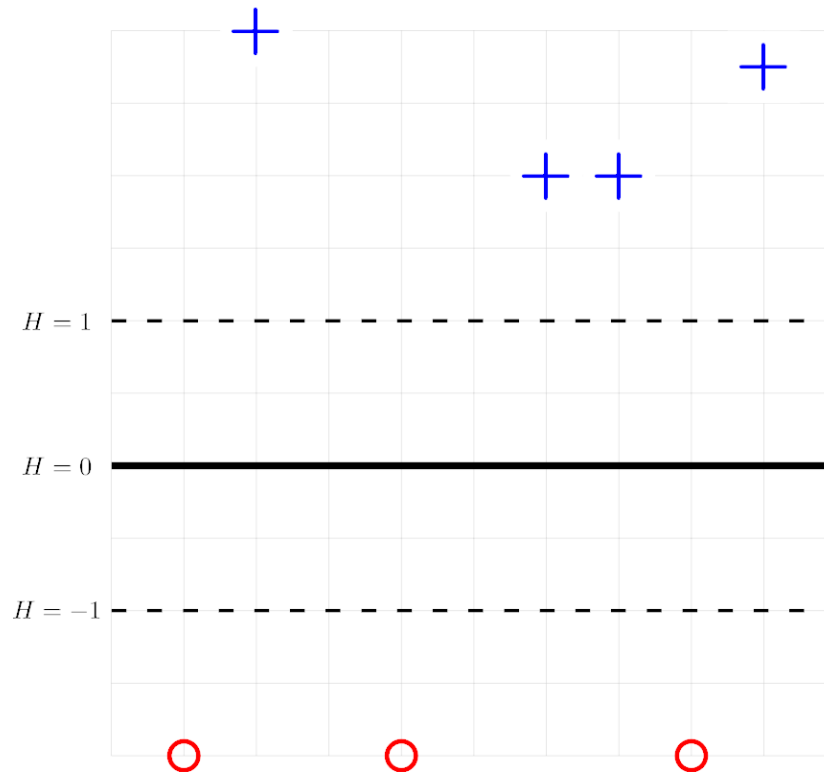
Solution

True.

Problem #089

Tags: SVMs, gradients

Consider the data set shown below. The points marked with "+" have label +1, while the "(\circ)" points have label -1. Shown are the places where a linear prediction function $\mathcal{H}$ is equal to zero (the thick solid line), 1, and -1 (the dotted lines). Each cell of the grid is 1 unit by 1 unit. The origin is not specified (it is not necessary to know the absolute coordinates of any points for this problem).



Let $(\vec{w}')$ be the parameter vector corresponding to prediction function (H) shown above. Let $(R(\vec{w}))$ be the risk with respect to the hinge loss on this data. True or False: the gradient of (R) evaluated at $(\vec{w}')$ is $(\vec{0})$.

True False

Solution

True.

Problem #090

Tags: gradients

Let $(\vec{x}, \vec{\mu})$ and $(\vec{b})$ be vectors in $(\mathbb{R}^n)$, and let (A) be a symmetric $(n \times n)$ matrix. Consider the function:

$$f(\vec{x}) = \vec{x}^T A \vec{x} + (\vec{x} - \vec{\mu}) \cdot \vec{b}$$

What is $(\frac{d}{d \vec{x}} f(\vec{x}))$?

$(2 A \vec{x} + \vec{b})$ Show Answer

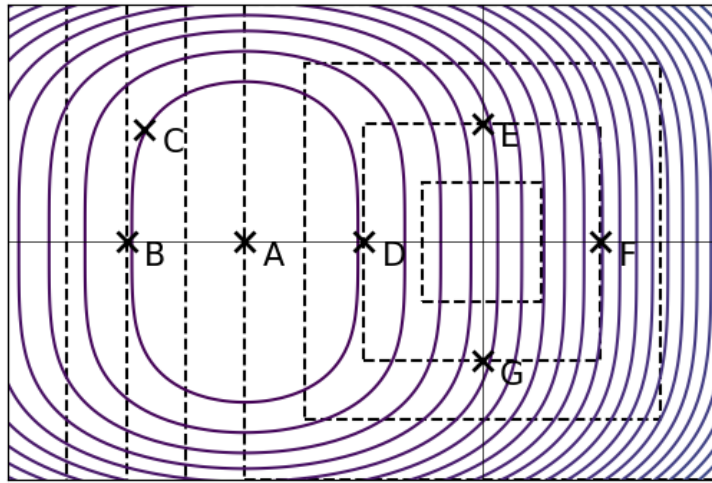
Problem #091

Tags: regularization

The " ℓ_1 " norm of a vector $(\vec{w} \in \mathbb{R}^d)$, written $\|\vec{w}\|_1$, is defined as:

$$\|\vec{w}\|_1 = \left(\sum_{i=1}^d |w_i| \right)$$

Let $R(\vec{w})$ be an unregularized risk function on a data set. The solid curves in the plot below are the contours of $R(\vec{w})$. The dashed lines show where $\|\vec{w}\|_1$ is equal to 1, 2, 3, and so on.



Let $\tilde{R}(\vec{w}) = R(\vec{w}) + \lambda \|\vec{w}\|_1$, with $(\lambda > 0)$. The point marked (A) is the minimizer of the unregularized risk. Suppose it is known that one of the other points is the minimizer of the regularized risk, $(\tilde{R}(\vec{w}))$, for some unknown $(\lambda > 0)$. Which point is it?

(B)

(C)

(D)

(E)

(F)

$\backslash(G\backslash)$

Solution

D.

Problem #092

Tags: kernel ridge regression

Suppose Gaussian kernel ridge regression is used to train a model on the following data set of $((x_i, y_i))$ pairs, using a kernel width parameter of $(\gamma = 1)$ and a regularization parameter of $(\lambda = 0)$:

i	x_i	y_i
1	1	3
2	2	4
3	3	3
4	6	-2
5	7	-5
6	8	-4
7	11	5
8	12	4
9	13	1

Let $(\vec{\alpha})$ be the solution to the dual problem. What will be the sign of (α_5) ?

Positive

Negative

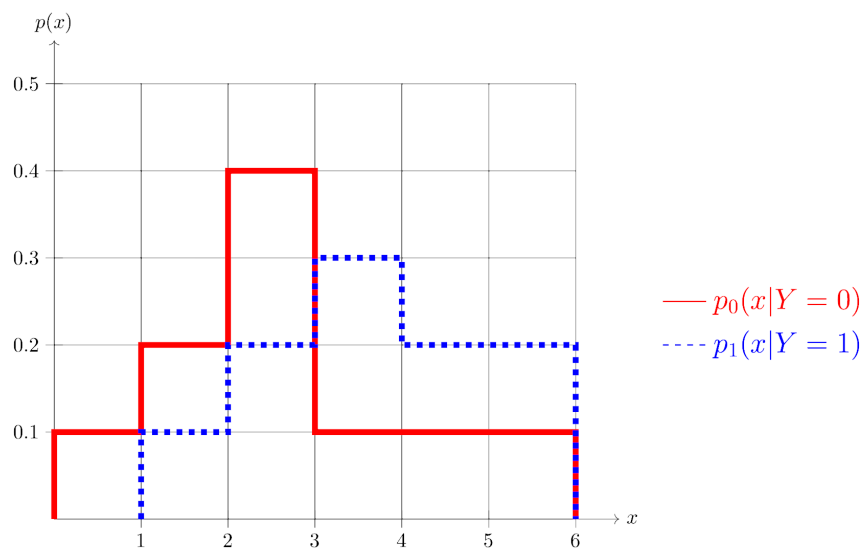
Solution

Negative. Video explanation: https://youtu.be/K_1cxeQAkdk

Problem #093

Tags: bayes error, bayes classifier

Shown below are two conditional densities, $p_1(x|Y=1)$ and $p_0(x|Y=0)$, describing the distribution of a continuous random variable X for two classes: $(Y=0)$ (the solid line) and $(Y=1)$ (the dashed line). You may assume that both densities are piecewise constant.



Part 1)

What is $\Pr(1 \leq X \leq 3 | Y=0)$?

0.6 Show Answer

Part 2)

Suppose $\Pr(Y=1) = \Pr(Y=0) = 0.5$. What is the prediction of the Bayes classifier at $x = 2.5$?

Class 0 (the solid class)

Class 1 (the dashed class)

Solution

Class 0.

Part 3)

Suppose again that $\Pr(Y=1) = \Pr(Y=0) = 0.5$. What is the Bayes error with respect to this distribution?

0.30 Show Answer

Part 4)

Now suppose $\Pr(Y = 1) = 0.7$ and $\Pr(Y = 0) = 0.3$. What is the prediction of the Bayes classifier at $x = 2.5$?

Class 0 (the solid class)

Class 1 (the dashed class)

Solution

Class 1.

Part 5)

Suppose again that $\Pr(Y = 1) = 0.7$ and $\Pr(Y = 0) = 0.3$. What is the Bayes error with respect to this distribution?

0.27 Show Answer

Solution

Video explanation: <https://youtu.be/mczKmVUJauI>

Problem #094

Tags: histogram estimators

In this problem, consider the following labeled data set of 15 points, 5 from Class 1 and 10 from Class 0.

x_i	y_i
0.2	1
1.4	0
2.2	1
3.1	0
3.3	1
4.7	0
4.8	1
5.1	0
5.2	0
5.5	1
5.7	0
6.1	0
6.3	0
6.8	0
7.9	0

Suppose the class conditional densities are estimated using a histogram estimator with bins: $\backslash([0, 3), [3, 5), [5, 6), \backslash)$ and $\backslash([6, 10)\backslash)$. Note that the bins are not all the same width!

In all of the parts below, you may write your answer either as a decimal or as a fraction.

Part 1)

What is the estimate of the Class 0 density at $(x = 3.5)$? That is, what is the estimate for $(p(3.5 \text{ given } Y = 0))$?

0.1 Show Answer

Part 2)

Using the same histogram estimator, what is the estimate of $(\text{pr}(Y = 1 \text{ given } x = 3.5))$?

0.5 Show Answer

Solution

Video explanation: <https://youtu.be/0WFYpsDapC8>

Problem #095

Tags: kernel ridge regression

Suppose a prediction function $(H(\text{vec } x))$ is trained using kernel ridge regression on the data below using regularization parameter $(\lambda = 4)$ and kernel $(\kappa(\text{vec } x, \text{vec } x') = (1 + \text{vec } x \cdot \text{vec } x')^2)$:

$(\text{vec } x\{1\} = (1, 2, 0))$, $(y_1 = 1)$, $(\text{vec } x\{2\} = (0, 1, 2))$, $(y_2 = -1)$, $(\text{vec } x\{3\} = (2, 0, 2))$, $(y_3 = 1)$, $(\text{vec } x\{4\} = (1, 0, 1))$, $(y_4 = -1)$, $(\text{vec } x\{5\} = (0, 0, 0))$, $(y_5 = 1)$ Suppose the solution to the dual problem is $(\text{vec } \alpha = (2, 0, 1, -1, 2))$.

Consider a new point $(\text{vec } x = (1, 1, 1))$. What is $(H(\text{vec } x))$?

50 Show Answer

Solution

Video explanation: <https://youtu.be/miyY9BeL0QI>

Problem #096

Tags: Gaussians

Recall that the density of a (d) -dimensional "axis-aligned" Gaussian (that is, a Gaussian with diagonal covariance matrix (C)) is given by:

$$f(x_1, \dots, x_d) = \frac{1}{(2\pi)^{d/2}} |C|^{-1/2} \exp\left\{-\frac{1}{2} \left(\frac{x_1 - \mu_1}{\sigma_1^2} + \dots + \frac{x_d - \mu_d}{\sigma_d^2}\right)^2\right\}$$

Consider the *marginal* density $(p_1(x_1))$, which is the density of the first coordinate, (x_1) , of a (d) -dimensional axis-aligned Gaussian.

True or False: $(p_1(x_1))$ must be a Gaussian density.

True False

Solution

True. Video explanation: <https://youtu.be/5ZJ6ZlvgMGk>

Problem #097

Tags: naive bayes

Consider the below data set collecting information on a set of 10 flowers.

#	Petals	Has Leaves?	Species
4		Yes	A
3		Yes	A
4		No	B
2		Yes	A
3		No	B
2		No	B
4		Yes	A
3		No	B
2		No	A
2		Yes	B

What species does a Naive Bayes classifier predict for a flower with 3 petals and no leaves?

A

B

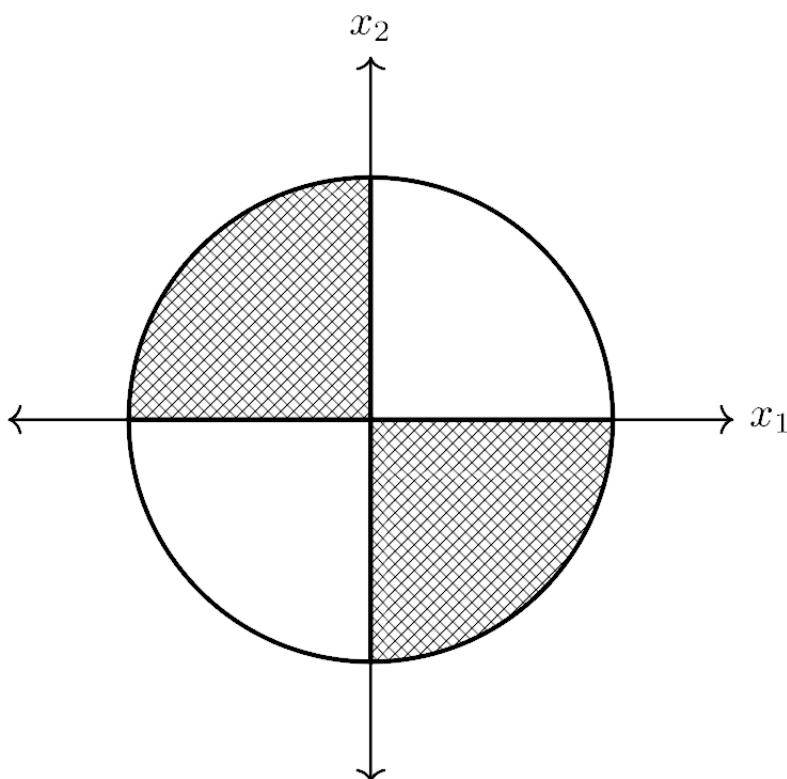
Solution

B. Video explanation: <https://youtu.be/iKPAhJJcu6s>

Problem #098

Tags: conditional independence

Suppose Justin has a dartboard at home that looks like the below:



Justin uses the dartboard to determine how long the midterm will be: if he throws a dart and it lands in the shaded region, the midterm will have 17 questions; if it lands in the unshaded region, it will have 16 questions.

Assume that Justin's dart throws are drawn from a uniform distribution on the dartboard, and that the dart always hits the board (that is, the density function is constant everywhere on the dartboard, and zero off of the dartboard). Let (X_1) be the horizontal component of a dart throw and (X_2) be the vertical component. Let (Q) be the number of questions on the exam; since it is chosen randomly, it is also a random number.

Part 1)

True or False: (X_1) and (X_2) are independent.

True False

Solution

False.

Part 2)

True or False: X_1 and X_2 are conditionally independent given Q .

True False

Solution

False.

Part 3)

True or False: X_1 and Q are independent.

True False

Solution

True.

Part 4)

True or False: X_1 and Q are conditionally independent given X_2 .

True False

Solution

False.

Solution

Video explanation: <https://youtu.be/b4XlZsePCgU>

Problem #099

Tags: Gaussians, linear and quadratic discriminant analysis

Suppose the underlying class-conditional densities in a binary classification problem are known to be multivariate Gaussians.

Suppose a Quadratic Discriminant Analysis (QDA) classifier using full covariance matrices for each class is trained on a data set of n points sampled from these densities.

True or False: As the size of the data set grows (that is, as $n \rightarrow \infty$), the training error of the QDA classifier must approach zero.

True False

Solution

False. Video explanation: <https://youtu.be/t40ex-JCYLY>

Problem #100

Tags: Gaussians, maximum likelihood

Suppose a univariate Gaussian density function $\hat{f}$ is fit to a set of data using the method of maximum likelihood estimation (MLE).

True or False: $\hat{f}$ must be between 0 and 1 everywhere. That is, it must be the case that for every $x \in \mathbb{R}$, $0 < \hat{f}(x) \leq 1$.

True False

Solution

False. Video explanation: <https://youtu.be/zvpLrG4FYEc>

Problem #101

Tags: linear and quadratic discriminant analysis

Suppose Quadratic Discriminant Analysis (QDA) is used to train a classifier on the following data set of $((x_i, y_i))$ pairs, where x_i is the feature and y_i is the class label:

x_i	y_i
0	0
2	0
4	0
1	1
3	1
5	1

Univariate Gaussians are used to model the class conditional densities, each with their own mean and variance.

What is the prediction of the QDA classifier at $(x = 2.25)$?

Class 0

Class 1

Solution

Class 0.

Video explanation: <https://youtu.be/5VxizVoBHsA>

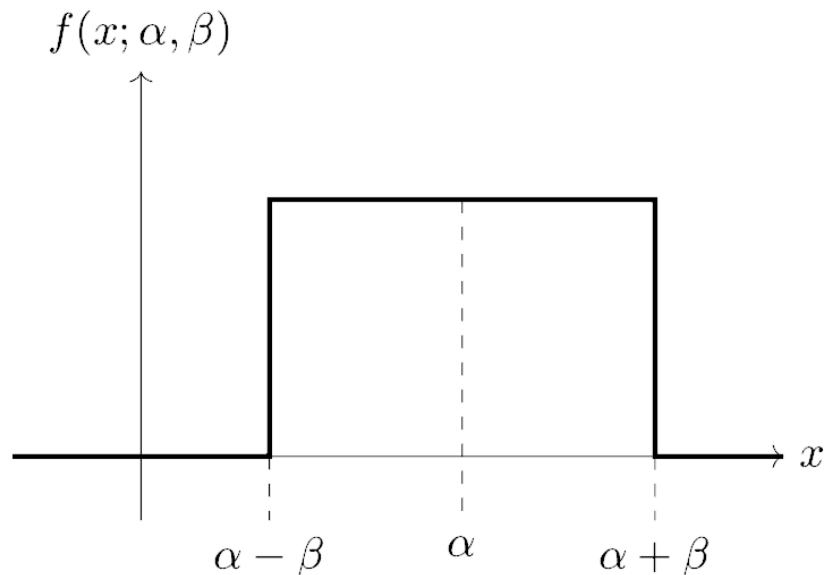
Problem #102

Tags: maximum likelihood

Consider *Justin's rectangle* density. It is a parametric density with two parameters, (α) and (β) , and pdf:

$$f(x; \alpha, \beta) = \begin{cases} \frac{1}{2\beta} & \text{if } x \in [\alpha - \beta, \alpha + \beta] \\ 0 & \text{otherwise} \end{cases}$$

A picture of the density is shown below for convenience:



In all of the below parts, let $(\mathcal{X} = \{1, 2, 3, 6, 9\})$ be a data set of 5 points generated from the rectangle density.

Your answers to the below problems should all be in the form of a number. You may leave your answer as an unsimplified fraction or a decimal, if you prefer.

Part 1)

Let $L(\alpha, \beta; \mathbf{X})$ be the likelihood function (with respect to the data given above). What is $L(6, 5)$? Note that L is the likelihood, *not* the *log*-likelihood.

10^{-5} Show Answer

Part 2)

What is $L(3, 2)$?

0 Show Answer

Part 3)

What are the maximum likelihood estimates of α and β ?

α : 5 Show Answer β : 4 Show Answer

Solution

Video explanation: <https://youtu.be/loc1xv2QNJk>

Problem #103

Tags: covariance, object type

Part 1)

Let $\mathbf{X}$ be a data set of n points in $\mathbb{R}^d$, and let $\vec{\alpha}$ be the solution to the kernel ridge regression dual problem. What type of object is $\vec{\alpha}$?

A d -dimensional vector

An n -dimensional vector

A $d \times n$ matrix

A scalar

Solution

B.

Part 2)

Suppose $\{\vec{x}_1, \dots, \vec{x}_n\}$ is a set of n points in $\mathbb{R}^d$, $\{y_1, \dots, y_n\}$ is a set of n labels (each either -1 or 1), $\vec{w}$ is a k -dimensional vector, and λ is a scalar.

Let $(\vec{\phi} : \mathbb{R}^d \rightarrow \mathbb{R}^k)$ be a feature map.

What type of object is the following?

$$\left[\sum_{i=1}^n \left(-\frac{1}{n} \lambda \vec{\phi}(\vec{x}_i) \cdot \vec{w} - y_i \right) \vec{\phi}(\vec{x}_i) \right]$$

A scalar

A d -dimensional vector

An n -dimensional vector

A k -dimensional vector

An $n \times n$ matrix

A $d \times d$ matrix

Solution

D.

Part 3)

Let $(\vec{x}_1, \dots, \vec{x}_n)$ be a set of n points in $\mathbb{R}^d$. Let $(\vec{\mu} = \sum_{i=1}^n \vec{x}_i)$ be the mean of the data set, and let (C) be the sample covariance matrix.

What type of object is the following?

$$\left[-\frac{1}{2} (\vec{x}_1 - \vec{\mu})^T C^{-1} (\vec{x}_1 - \vec{\mu}) \right]$$

A scalar

A d -dimensional vector

An n -dimensional vector

An $n \times n$ matrix

A $d \times d$ matrix

Solution

A.

Part 4)

Let $(\vec{x}_1, \dots, \vec{x}_n)$ be a data set of n points in $\mathbb{R}^d$ sampled from a multivariate Gaussian with known covariance matrix but unknown mean, $(\vec{\mu})$. Let $(\mathcal{L}(\vec{\mu}))$ be the likelihood function for the Gaussian's mean, $(\vec{\mu})$. What type of object is $(\mathcal{L})$?

A function from $(\mathbb{R})$ to $(\mathbb{R})$

A function from $\mathbb{R}^d$ to $\mathbb{R}$

A function from $\mathbb{R}^n$ to $\mathbb{R}$

A function from $\mathbb{R}^d$ to $\mathbb{R}^n$

Solution

Video explanation: <https://youtu.be/wr8sNCEiIQs>

Problem #104

Tags: histogram estimators

Let $((\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n))$ be a set of n points in a binary classification problem, where $\mathbf{x}_i \in \mathbb{R}^2$ and $y_i \in \{0, 1\}$.

Suppose a classifier is trained by estimating the class-conditional densities with histograms using rectangular bins and applying the Bayes classification rule.

True or False: it is always possible to achieve a 100% training accuracy with this classifier by choosing the rectangular bins to be sufficiently small. You may assume that no two points $\mathbf{x}_i$ and $\mathbf{x}_j$ are identical.

True False

Solution

True. Video explanation: <https://youtu.be/A1fBjOnjs5E>

Problem #105

Tags: naive bayes

Does standardizing the data possibly change the prediction made by Gaussian Naive Bayes?

More precisely, let $((\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n))$ be a set of training data for a binary classification problem, where each $\mathbf{x}_i \in \mathbb{R}^d$ and each $y_i \in \{0, 1\}$. Suppose that when a Gaussian Naive Bayes classifier is trained on this data set, $\mathbf{x}_1$ is predicted to be from Class (0) .

Now suppose that the data set is standardized by subtracting the mean of each feature and dividing by its standard deviation. This is done for the data set as a whole, not separately for each class. Let $(\mathbf{z}_1, \dots, \mathbf{z}_n)$ be the standardized data set.

True or False: when a Gaussian Naive Bayes classifier is trained on the standardized data set, it must still predict $\mathbf{x}_1$ to be from Class (0) .

True False

Solution

True.

Video explanation: https://youtu.be/hGIWsEk_tYI

Problem #106

Tags: covariance, maximum likelihood

Consider the following set of 6 data points:

$$\begin{aligned} \vec{x}_1 &= (1, 2)^T & \vec{x}_2 &= (-1, 1)^T \\ \vec{x}_3 &= (1, 3)^T & \vec{x}_4 &= (1, 4)^T \\ \vec{x}_5 &= (-1, 1)^T & \vec{x}_6 &= (-1, 1)^T \end{aligned}$$

In the below parts, your answers should be given as numbers. You may leave your answer as an unsimplified fraction or a decimal, if you prefer.

Part 1)

What is the (1,2) entry of the sample covariance matrix?

1 Show Answer

Part 2)

What is the (2,2) entry of the sample covariance matrix?

8/6 Show Answer

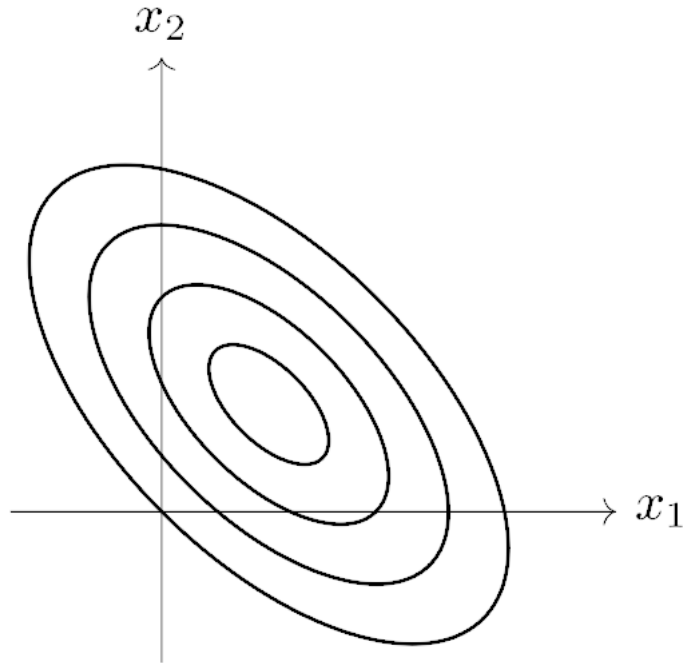
Solution

Video explanation: <https://youtu.be/BvFKfpGVR9k>

Problem #107

Tags: covariance, Gaussians

The picture below shows the contours of a multivariate Gaussian density function:



Which one of the following could possibly be the covariance matrix of this Gaussian?

$$\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$$

$$\begin{pmatrix} -3 & 1 \\ 1 & -3 \end{pmatrix}$$

$$\begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 3 & -1 \\ -1 & 1 \end{pmatrix}$$

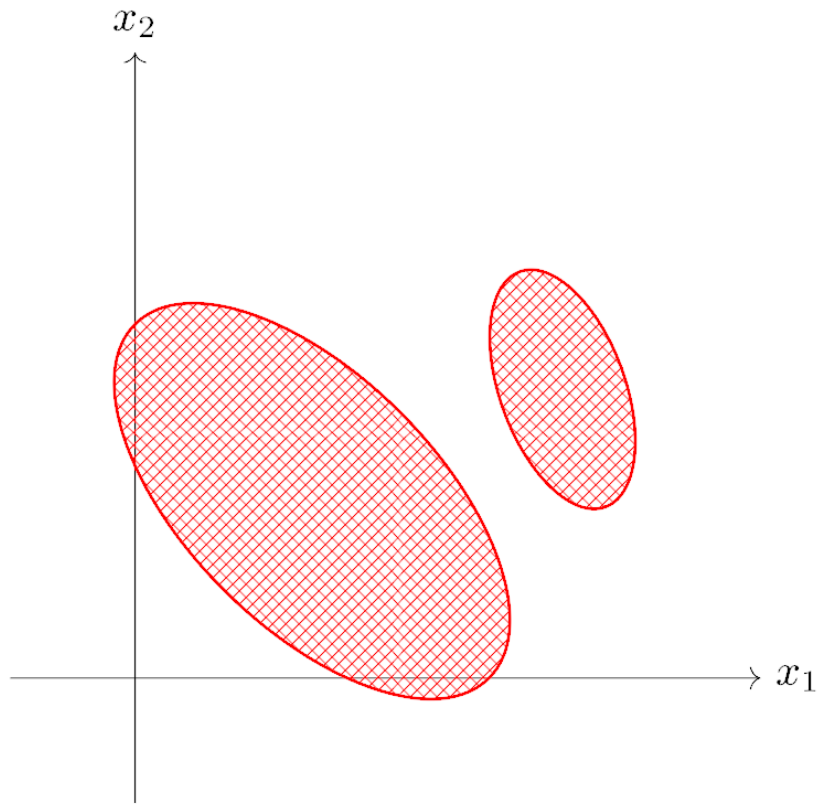
Solution

C. Video explanation: <https://youtu.be/5b1nzF0yYeE>

Problem #108

Tags: linear and quadratic discriminant analysis

The picture below shows the decision boundary of a binary classifier. The shaded region is where the classifier predicts for Class 1; everywhere else, the classifier predicts for Class 0.



True or False: this could be the decision boundary of a Quadratic Discriminant Analysis (QDA) classifier that models the class-conditional densities as multivariate Gaussians with full covariance matrices.

True False

Solution

False. Video explanation: https://youtu.be/fQm3_JcpVys

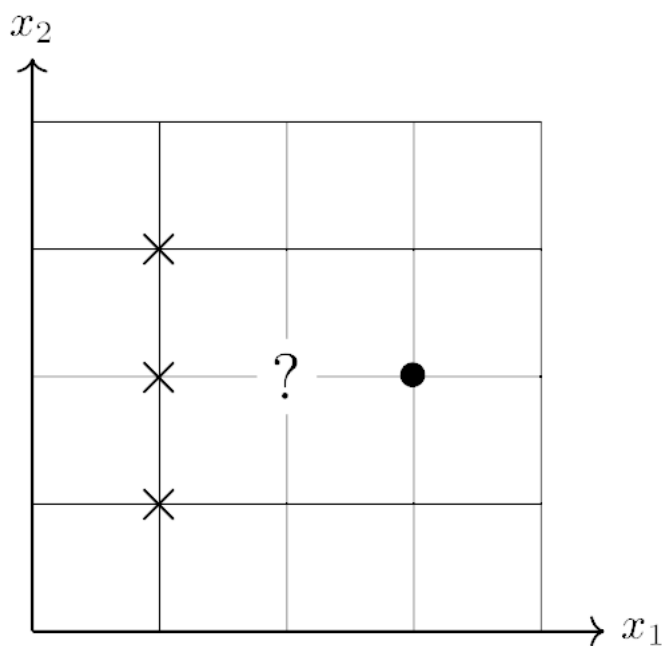
Problem #109

Tags: kernel ridge regression

Consider the following data set of four points whose feature vectors are in $\mathbb{R}^2$ and whose labels are in $\{-1, 1\}$:

i	1	2	3	4
$\vec{x}^{(i)}$	(1,1)	(1,2)	(1,3)	(3,2)
y_i	1	1	1	-1

For convenience, we've plotted the data below. Each point is labeled with either the positive class (denoted by $\times$) or the negative class (denoted by $\bullet$).



Suppose an unnamed kernel classifier $(H(\vec{x}) = \sum_{i=1}^n \alpha_i \kappa(\vec{x}, \vec{x}_i))$ has been trained on this data using a (spherical) Gaussian kernel and kernel width parameter $(\gamma = 1)$. Suppose the solution to the dual problem is found to be $(\vec{\alpha} = (1, 1, 1, -3)^T)$.

What class will the classifier predict for the point $((2,2))$? For convenience, we've plotted this point on the graph above as a question mark.

+1 (the $\times$ class)

-1 (the $\bullet$ class)

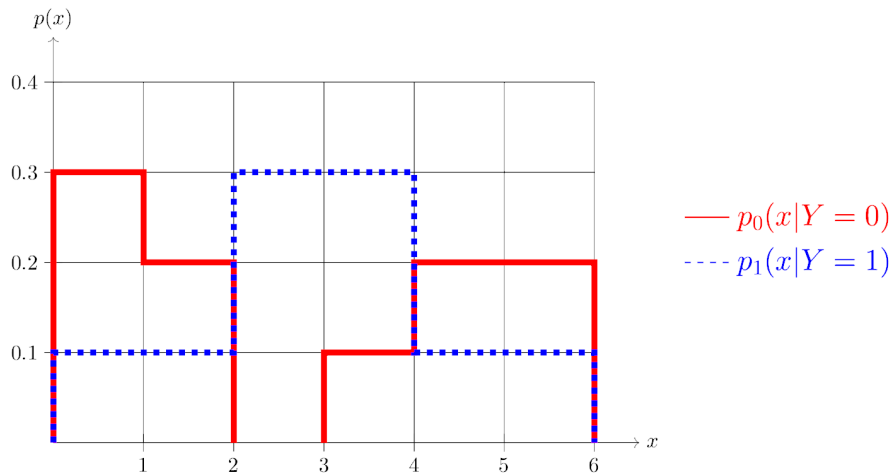
Solution

-1 (the \(\bullet\) class).

Problem #110

Tags: bayes error, bayes classifier

Shown below are two conditional densities, $p_1(x|Y=1)$ and $p_0(x|Y=0)$, describing the distribution of a continuous random variable X for two classes: $(Y=0)$ (the solid line) and $(Y=1)$ (the dashed line). You may assume that both densities are piecewise constant.



Part 1)

Suppose that $\Pr(Y=1) = \Pr(Y=0) = 0.5$. What is $\Pr(1 \leq X \leq 3)$?

0.3 Show Answer

Part 2)

Suppose $\Pr(Y=1) = \Pr(Y=0) = 0.5$. What is the prediction of the Bayes classifier at $(x=1.5)$?

Class 0 (the solid class)

Class 1 (the dashed class)

Solution

Class 0.

Part 3)

Suppose again that $\Pr(Y = 1) = \Pr(Y = 0) = 0.5$. What is the Bayes error with respect to this distribution?

0.25 Show Answer

Part 4)

Now suppose $\Pr(Y = 1) = 0.7$ and $\Pr(Y = 0) = 0.3$. What is $\Pr(1 \leq X \leq 3)$?

0.34 Show Answer

Part 5)

Suppose again that $\Pr(Y = 1) = 0.7$ and $\Pr(Y = 0) = 0.3$. What is the prediction of the Bayes classifier at $x = 1.5$?

Class 0 (the solid class)

Class 1 (the dashed class)

Solution

Class 1.

Part 6)

Suppose again that $\Pr(Y = 1) = 0.7$ and $\Pr(Y = 0) = 0.3$. What is the Bayes error with respect to this distribution?

0.28 Show Answer

Problem #111

Tags: bayes classifier, histogram estimators

In this problem, consider the following labeled data set of 19 points, 7 from Class 1 and 12 from Class 0.

x_i	y_i
0.01	0
0.08	0
0.11	1
0.13	0
0.19	1
0.22	1
0.23	0
0.31	0
0.33	1
0.42	1
0.47	0
0.48	1
0.51	0
0.52	0
0.55	1
0.57	0
0.61	0
0.63	0
0.79	0

Suppose the class conditional densities are estimated using a histogram estimator with bins: $\backslash([0, .25), [.25, .5), [.5, .75), \backslash)$ and $\backslash([.75, 1.0)\backslash)$.

In all of the parts below, you may write your answer either as a decimal or as a fraction.

Part 1)

What is the estimate of the Class 0 density at $(x = 0.6)$? That is, what is the estimate $(\hat{p}(0.6 \text{ given } Y = 0))$?

5/3 Show Answer

Part 2)

Using the same histogram estimator, what is the estimate $(\hat{\text{pr}}(Y = 1 \text{ given } x = 0.35))$?

3/5 = 0.6 Show Answer

Part 3)

What is the estimate of the marginal density of (x) at $(x = 0.1)$? That is, what is $(\hat{p}(0.1))$?

7/(19*0.25) = 1.47 Show Answer

Part 4)

Let $(\hat{p}(x \text{ given } Y = 0))$ be the histogram density estimate for the Class 0 conditional density. What is

$(\int_0^1 \hat{p}(x \text{ given } Y = 0) \, dx)$?

1 Show Answer

Problem #112

Tags: kernel ridge regression

Suppose a prediction function $(H(\vec{x}))$ is trained using kernel ridge regression on the data below using regularization parameter $(\lambda = 4)$ and kernel $(\kappa(\vec{x}, \vec{x}') = (1 + \vec{x} \cdot \vec{x}')^2)$:

$(\vec{x}_1 = (0, 1, 1))$, $(y_1 = 1)$, $(\vec{x}_2 = (1, 1, 1))$, $(y_2 = -1)$, $(\vec{x}_3 = (2, 2, 2))$, $(y_3 = 1)$, $(\vec{x}_4 = (1, 1, 0))$, $(y_4 = -1)$, $(\vec{x}_5 = (0, 1, 0))$, $(y_5 = 1)$ Suppose the solution to the dual problem is $(\vec{\alpha} = (-1, 1, 0, 3, -2))$.

Consider a new point $(\vec{x} = (2, 0, 1)^T)$. What is $(H(\vec{x}))$?

37 Show Answer

Problem #113

Tags: covariance, maximum likelihood

Consider the following set of 6 data points:

$$\begin{aligned} \vec{x}_1 &= (2, 4, 4)^T \\ \vec{x}_2 &= (-1, 2, 1)^T \\ \vec{x}_3 &= (3, -3, 2)^T \\ \vec{x}_4 &= (0, -3, -3)^T \end{aligned}$$

In the below parts, your answers should be given as numbers. You may leave your answer as an unsimplified fraction or a decimal, if you prefer.

Part 1)

What is the (1,3) entry of the sample covariance matrix?

9/4 Show Answer

Part 2)

What is the (1,2) entry of the sample covariance matrix?

-3/4 Show Answer

Problem #114

Tags: conditional independence, Gaussians, covariance

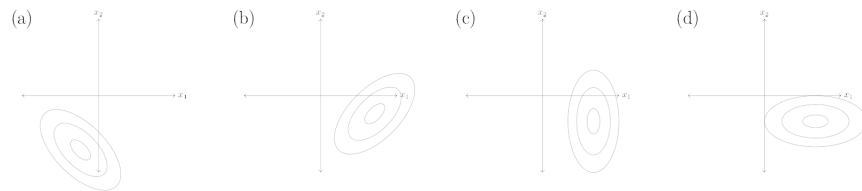
Let X_1 and X_2 be two independent random variables. Suppose the distribution of X_1 has the Gaussian density:

$$p_1(x_1) = \frac{1}{\sqrt{8\pi}} e^{-\frac{(x_1 - 2)^2}{8}},$$

while the distribution of X_2 has the Gaussian density:

$$p_2(x_2) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x_2 + 1)^2}{2}}.$$

Which one of the following pictures shows the contours of the joint density $p(x_1, x_2)$ (the density for the joint distribution of X_1 and X_2)?



Picture (a)

Picture (b)

Picture (c)

Picture (d)

Solution

Picture (d).

Problem #115

Tags: Gaussians, maximum likelihood

Suppose it is known that the distribution of a random variable (X) has a univariate Gaussian density function (f) .

True or False: (f) must be between 0 and 1 everywhere. That is, it must be the case that for every $(x \in \mathbb{R})$, $(0 < f(x) \leq 1)$.

True False

Solution

False.

Problem #116

Tags: covariance, Gaussians, bayes error

Suppose that, in a binary classification setting, the true underlying class-conditional densities $(p(\vec{x} \text{ given } Y=0))$ and $(p(\vec{x} \text{ given } Y=1))$ are known to each be multivariate Gaussians with full covariance matrices. Suppose, also, that $(\text{pr}(Y = 1) = \text{pr}(Y = 0) = \frac{1}{2})$.

True or False: it is possible that the Bayes error in this case is exactly zero.

True False

Solution

False.

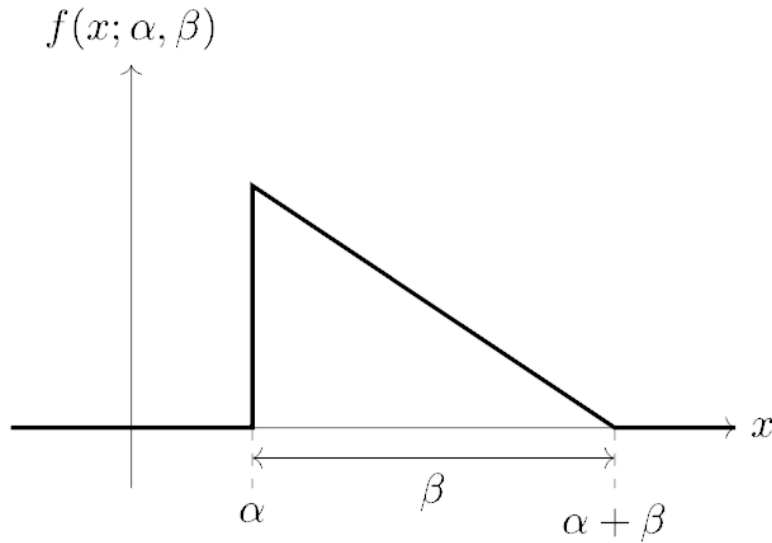
Problem #117

Tags: maximum likelihood

Consider *Justin's right-triangle* density. It is a parametric density with two parameters, (α) and (β) (with $(\beta > 0)$), and pdf:

$$f(x; \alpha, \beta) = \begin{cases} \frac{2}{\beta} (x - \alpha) & \text{if } x \in [\alpha, \alpha + \beta] \\ 0 & \text{otherwise} \end{cases}$$

A picture of the density is shown below for convenience:



Your answers to the below problems should all be in the form of a number. You may leave your answer as an unsimplified fraction or a decimal, if you prefer.

Part 1)

Let $X = \{0, 1, 2\}$ be a data set of 4 points and let $L(\alpha, \beta; X)$ be the likelihood function (with respect to this data). What is $L(0, 2)$? Note that L is the likelihood, *not* the *log*-likelihood.

0 Show Answer

Part 2)

Let $X = \{0, 1, 2\}$ be the same data set as in the previous part. What is $L(0, 4)$?

$3/64 = 0.0468$ Show Answer

Part 3)

Now let $X = \{6, 7\}$ be a new data set of just two points. What are the maximum likelihood estimates of α and β with respect to this data? Remember, $\beta > 0$.

α : 6 Show Answer β : 3 Show Answer

Problem #118

Tags: object type

Part 1)

Let $\mathcal{X}$ be a data set of n points in $\mathbb{R}^d$, and let C be the sample covariance matrix of $\mathcal{X}$. Let $|C|$ denote the determinant of C . What type of object is the following?

$$\frac{1}{(2\pi)^{d/2}} |C|^{-1/2}$$

A scalar

A d -dimensional vector

An n -dimensional vector

A $d \times n$ matrix

A $d \times d$ matrix

An $n \times n$ matrix

Solution

A.

Part 2)

Suppose $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ is a set of n points in $\mathbb{R}^d$, λ is a positive real number, $\mathbf{y} \in \mathbb{R}^n$ is a vector of targets, and $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^k$ is a feature map. Let K be the kernel matrix for this feature map on this data set, and let I be an identity matrix (the same shape as K). What type of object is the following?

$$\frac{1}{(K + n\lambda I)^{-1}} \mathbf{y}$$

A scalar

A d -dimensional vector

An n -dimensional vector

An $n \times n$ matrix

A $d \times d$ matrix

Solution

C.

Part 3)

Let $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ be a set of n points in $\mathbb{R}^d$. Let $\mu = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i$ be the mean of the data set, and let C be the sample covariance matrix.

What type of object is the following?

$$\frac{1}{C}(\|\vec{x}_1 - \vec{\mu}\|)$$

A scalar

A d -dimensional vector

An n -dimensional vector

An $n \times n$ matrix

A $d \times d$ matrix

Solution

B.

Part 4)

Let $\{\vec{x}_1, \dots, \vec{x}_n\}$ be a data set of n points, let $\alpha \in \mathbb{R}^n$, and let κ be a kernel function for a feature map $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^k$. Suppose also that $\vec{z} \in \mathbb{R}^d$. What type of object is the following?

$$\sum_{i=1}^n \alpha_i \kappa(\vec{x}_i, \vec{z})$$

A scalar

A d -dimensional vector

An n -dimensional vector

An $n \times n$ matrix

A $d \times d$ matrix

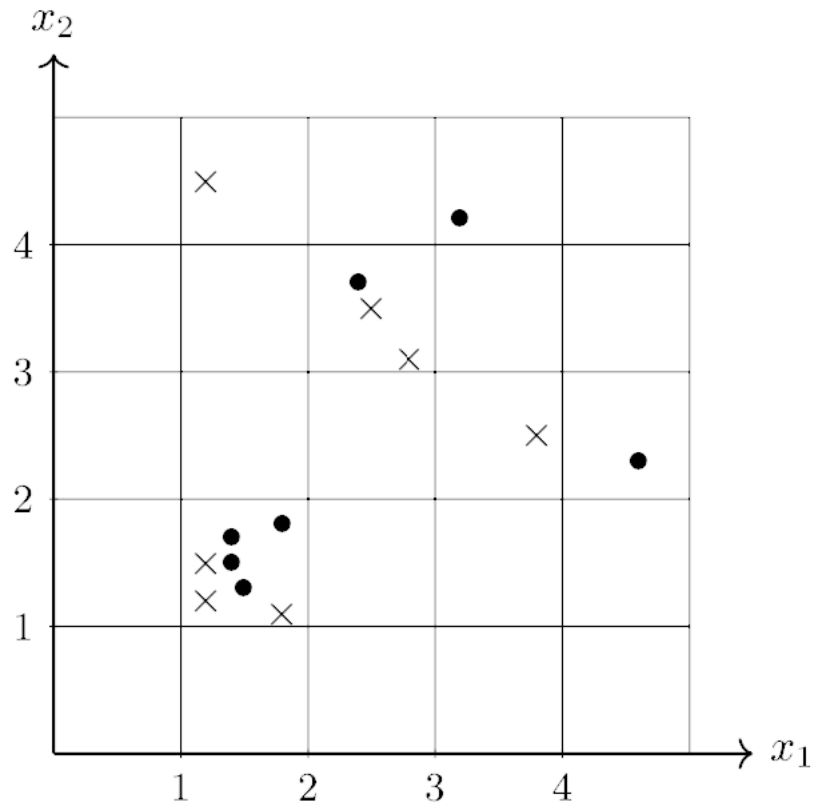
Solution

A.

Problem #119

Tags: histogram estimators

Consider the following data set of 14 points in $\mathbb{R}^2$. Each point has a label in $\{1, -1\}$. Points from Class 1 are marked with $\times$ and points from Class -1 are marked with $\bullet$.



Suppose the class-conditional density estimates are computed from this data using a histogram density estimator using the (1×1) bins shown in the figure above.

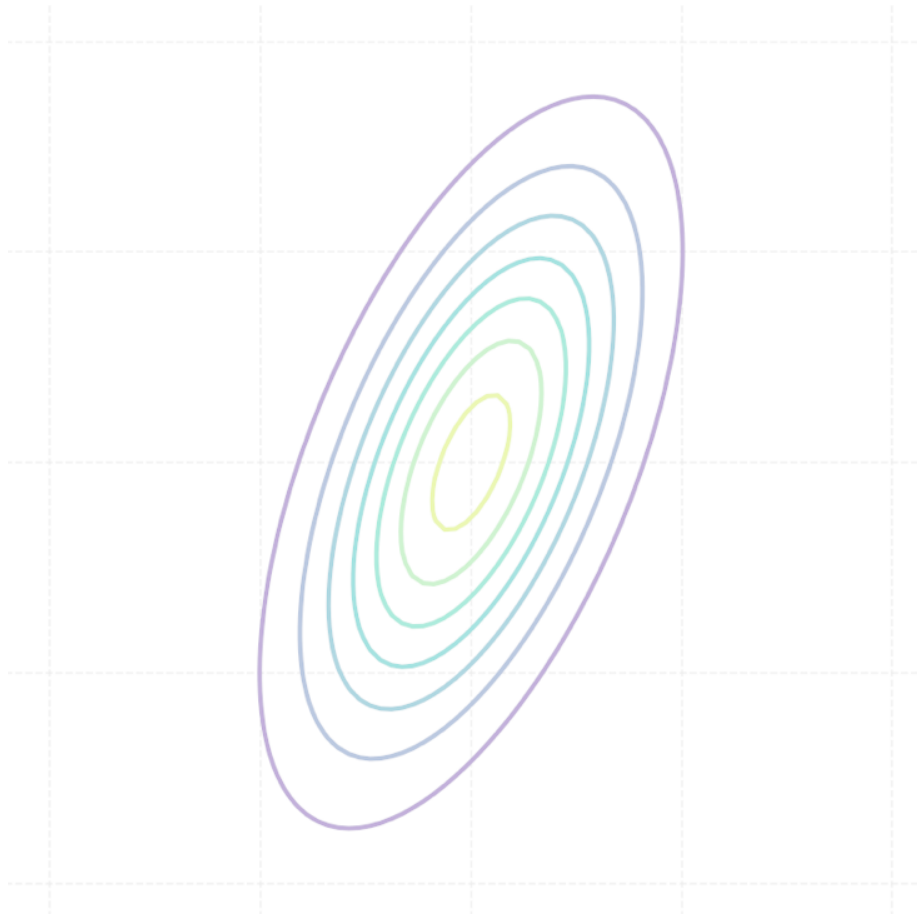
If these estimates are used in place of the true class-conditional densities in the Bayes classifier, what will be the training error of the classifier? That is, what percentage of the data above will be misclassified? You may leave your answer as a fraction or a decimal.

$4/14 = .2857$ Show Answer

Problem #120

Tags: covariance, Gaussians

The picture below shows the contours of a multivariate Gaussian density function:



Which one of the following could possibly be the covariance matrix of this Gaussian?

$$\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 \\ 1 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 3 & -1 \\ -1 & 1 \end{pmatrix}$$

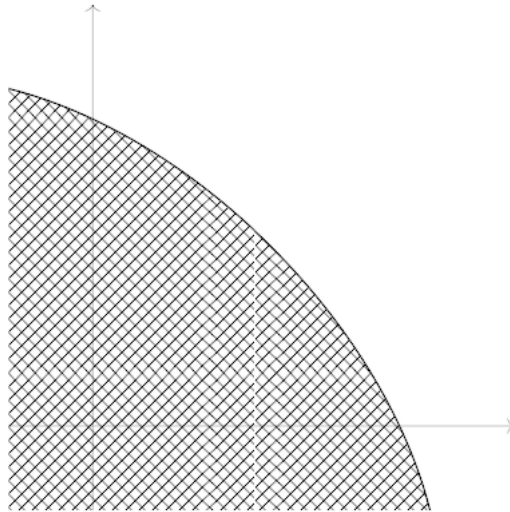
Solution

C.

Problem #121

Tags: linear and quadratic discriminant analysis

The picture below shows the decision boundary of a binary classifier. The shaded region is where the classifier predicts for Class 1; everywhere else, the classifier predicts for Class 0. You can assume that the shaded region extends infinitely to the left and down.



True or False: this could be the decision boundary of a classifier that estimates each class conditional density using two Gaussians with different means but the same, shared full covariance matrix, and applies the Bayes classification rule.

True False

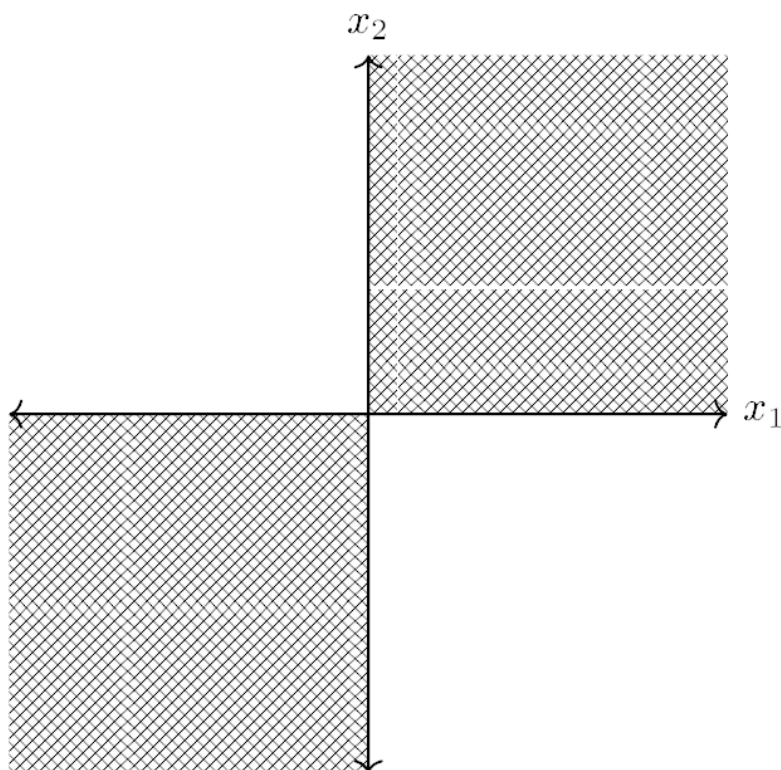
Solution

False.

Problem #122

Tags: conditional independence

Suppose that Justin has a wall at home that he has painted to look like the below:



Justin uses the wall to determine how long Redemption Midterm 02 will be: if he throws a dart at the origin and it lands in the shaded region, the redemption midterm will have 14 questions; if it lands in the unshaded region, it will have 2 questions.

Assume that Justin's dart throws are drawn from a spherical Gaussian whose mean is at the origin. Let (X_1) be the horizontal component of a dart throw and (X_2) be the vertical component. Let (Q) be the number of questions on the exam; since it is chosen randomly, it is also a random number. You can assume that the wall is infinitely large, and that the shaded regions extend infinitely up and to the right, and down and to the left.

Part 1)

True or False: (X_1) and (X_2) are independent.

True False

Solution

True.

Part 2)

True or False: (X_1) and (X_2) are conditionally independent given (Q) .

True False

Solution

False.

Part 3)

Let $(D = \sqrt{X_1^2 + X_2^2})$, and note that (D) is the distance of a dart throw from the origin. True or False: (D) and (Q) are independent.

True False

Solution

True.

Part 4)

True or False: (X_1) and (X_2) are conditionally independent given (D) .

True False

Solution

False.